

# Companion to Bird & de Moor, Algebra of Programming

## §9. Dynamic Programming

### §9.1. Theory

#### §9.1.1. Example: segmenting a list

- $(\mathsf{T}) = \text{concat}$   
 $\mathsf{T} = [\text{nil}, \text{cat}]$  takes a non-empty first list, so  $\mathsf{T}^\circ$  cuts a non-empty prefix off a list every way, and folding with  $\mathsf{T}$  flattens a list of segments (9.1.1a)
- $(\mathsf{h}) = \text{list}(\text{sum})$   
 $\mathsf{h} = [\text{nil}, \text{cons}(\text{sum} \times 1)]$  puts a segment's sum in front, so folding with  $\mathsf{h}$  sums every segment
- $\mathsf{H} = \text{partition list}(\text{sum})$   
 $\mathsf{H}$  segments a list every way, then sums each segment; with  $\mathsf{R}$  comparing the largest entry,  $\mathsf{M}$  cuts the list so that the largest segment sum is as small as possible

(9.1.1b)

|   | <b>x</b>                   | <b>h(x)</b>  |
|---|----------------------------|--------------|
| 1 | $\text{inl}(() )$          | $[\ ]$       |
| 2 | $\text{inr}([c], [ ])$     | $[c]$        |
| 3 | $\text{inr}([a, b], [c])$  | $[a + b, c]$ |
| 4 | $\text{inr}([a], [b + c])$ | $[a, b + c]$ |

- $\mathsf{H}([a, b, c], \mathsf{ys}) \iff \mathsf{ys} = [a + b + c] \vee \mathsf{ys} = [a, b + c] \vee \mathsf{ys} = [a + b, c] \vee \mathsf{ys} = [a, b, c]$  (9.1.1c)  
 $\mathsf{H}$  sends  $[a, b, c]$  to the sums of its four segmentations and nothing else
- $\mathsf{T}^\circ \mathsf{F}(\mathsf{H}) \mathsf{h} = \mathsf{H}$   
cut once, solve the rest, put the pieces back together

(9.1.1d)

$\mathsf{h} : \mathsf{FB} \rightarrow \mathsf{B}$  a map,  $\mathsf{T} : \mathsf{FA} \rightarrow \mathsf{A}$  an F-algebra,  $\mathsf{R} : \mathsf{B} \rightarrow \mathsf{B}$ .

$\mathsf{H} \triangleq (\mathsf{T})^\circ (\mathsf{h}) : \mathsf{A} \rightarrow \mathsf{B}$ ,  $\mathsf{M} \triangleq \frac{\mathsf{H}}{\mathsf{S}} \text{est}(\mathsf{R})$  the problem to be solved,  $(\mu \mathsf{X} : \mathsf{G}(\mathsf{X}))$  as in  $\triangleq \mu$ .

$\mathsf{DPSetting}(\mathsf{h}, \mathsf{R}) \triangleq \text{Map}(\mathsf{h}) \wedge \text{MonoAlg}(\mathsf{h}, \mathsf{R}^\circ) \wedge \mathsf{R}^\circ \mathsf{R}^\circ \subseteq \mathsf{R}^\circ$

$\mathsf{h}$  is a map,  $\mathsf{h}$  is monotonic on  $\mathsf{R}^\circ$ , and  $\mathsf{R}^\circ$  is transitive — assumed by every theorem below

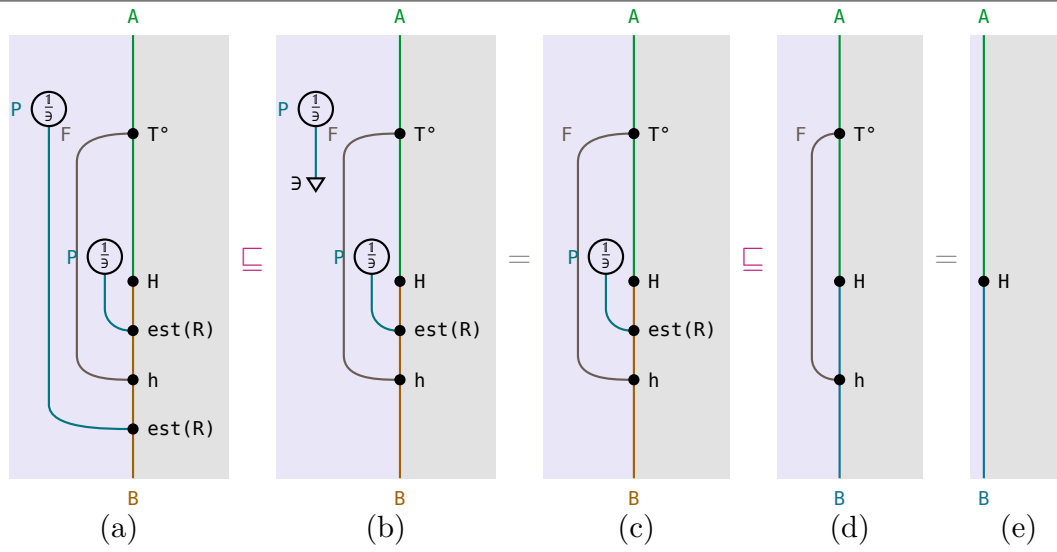
**$(\mu \mathsf{X} : \frac{\mathsf{T}^\circ}{\mathsf{S}} \mathsf{P}(\mathsf{F}(\mathsf{X}) \mathsf{h}) \text{est}(\mathsf{R})) \subseteq \frac{\mathsf{H}}{\mathsf{S}} \text{est}(\mathsf{R})$**   
every answer the recursion returns is an optimal one

Theorem 9.1

- $\frac{\mathsf{T}^\circ}{\mathsf{S}}$  takes the input apart one step every way;  $\mathsf{F}(\mathsf{X})$  solves each part by the recursion  $\mathsf{X}$ ;  $\mathsf{h}$  assembles each candidate;  $\text{est}(\mathsf{R})$  keeps a best one
- $\sqsubseteq \mathsf{M}$ : that never misses the optimum, though all candidates are never listed as  $\mathsf{H}$  would
- $\mathsf{h}$  monotonic on  $\mathsf{R}$ : a better part never makes the whole worse
- $\sqsubseteq$ , not  $=$ : some optima may not be returned; that the recursion returns anything at all needs  $\mathsf{T}^\circ$  to stop taking apart — dynamic programming theorem

$$T^\circ F(H) h = H \implies \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R) \sqsubseteq H \quad (9.2)$$

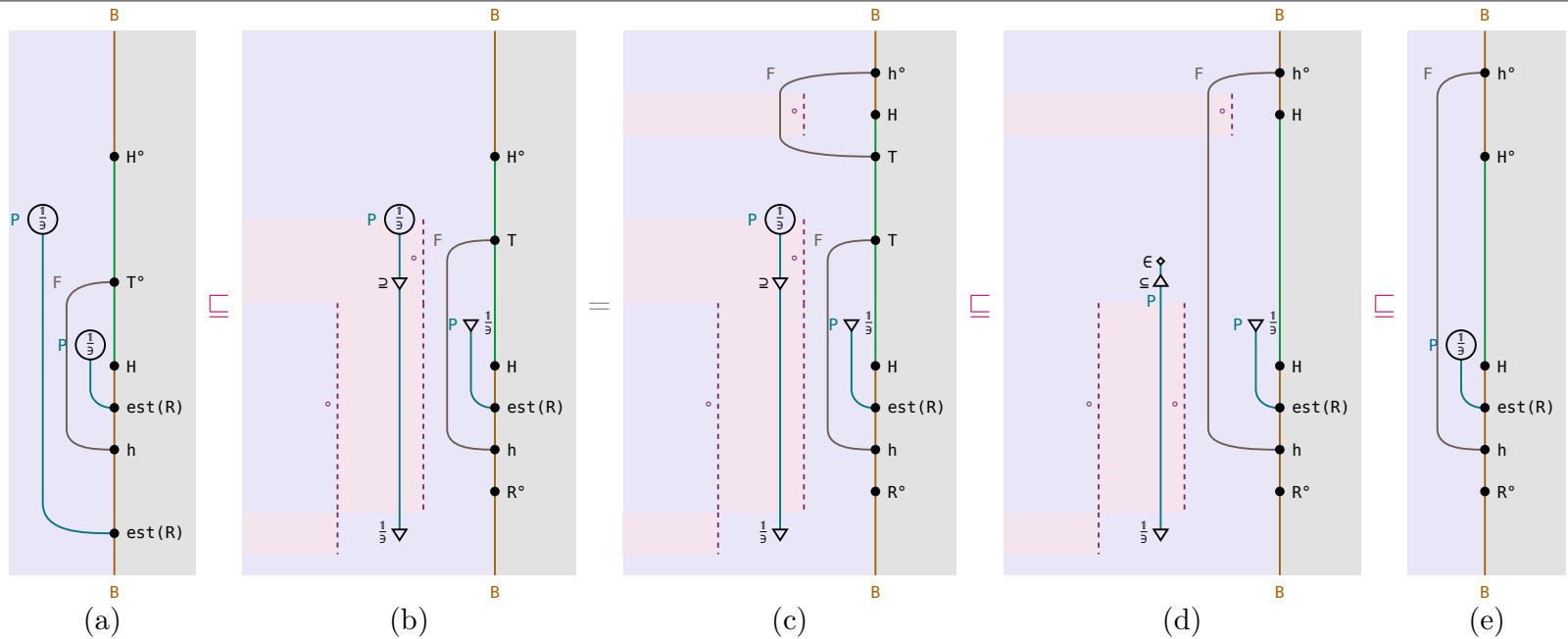
taking the input apart every way  $T$  allows ( $\frac{T^\circ}{\exists}$ ), solving each part by  $M$  and keeping an optimum ( $P(F(M)h) \text{est}(R)$ )  
returns only what  $H$  returns



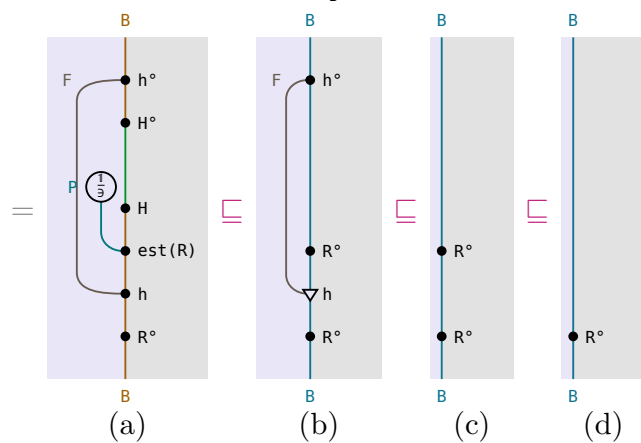
|     |                                                                                    |                                                 |
|-----|------------------------------------------------------------------------------------|-------------------------------------------------|
| (a) | $\frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R)$    | $P(S) \text{est}(R) \sqsubseteq \exists S$      |
| (b) | $\sqsubseteq \frac{T^\circ}{\exists} \exists F(\frac{H}{\exists} \text{est}(R)) h$ | $\frac{R}{\exists} \exists = R$                 |
| (c) | $= T^\circ F(\frac{H}{\exists} \text{est}(R)) h$                                   | $\frac{R}{\exists} \text{est}(R) \sqsubseteq S$ |
| (d) | $\sqsubseteq T^\circ F(H) h$                                                       | $T^\circ F(H) h = H$                            |
| (e) | $= H$                                                                              |                                                 |

$$T^\circ F(H) h = H \implies H^\circ \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R) \sqsubseteq R^\circ \quad (9.3)$$

for every  $\mathbf{b}$  that  $\mathbf{H}$  returns from an input, the step  $\frac{T^\circ}{\exists} P(F(\mathbf{M})h) \text{est}(R)$  returns from that input only  $\mathbf{b}'$  with  $\mathbf{R}$  relating  $\mathbf{b}'$  to  $\mathbf{b}$



|     |                                                                                                                      |                                                                   |
|-----|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| (a) | $H^\circ \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R)$                              | $P(S) \text{est}(R) \sqsubseteq \epsilon \setminus (S R^\circ)$   |
| (b) | $\sqsubseteq H^\circ \frac{T^\circ}{\exists} (\epsilon \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$    | $T^\circ F(H) h = H$<br>$\implies H^\circ = h^\circ F(H^\circ) T$ |
| (c) | $= h^\circ F(H^\circ) T \frac{T^\circ}{\exists} (\epsilon \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$ | $T \frac{T^\circ}{\exists} \sqsubseteq \epsilon$                  |
| (d) | $\sqsubseteq h^\circ F(H^\circ) \epsilon \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ)$                   | $S (S \setminus R) \sqsubseteq R$                                 |
| (e) | $\sqsubseteq h^\circ F(H^\circ) F(\frac{H}{\exists} \text{est}(R)) h R^\circ$                                        | $t(f g) = t(f) t(g)$                                              |



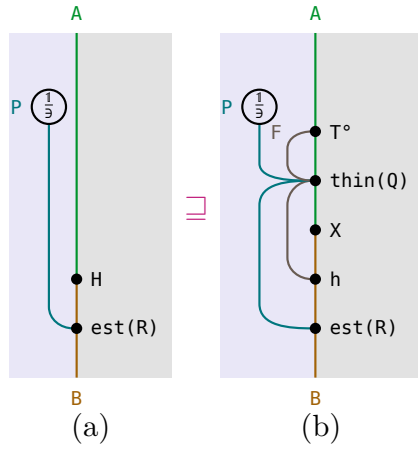
|     |                                                                  |                                                               |
|-----|------------------------------------------------------------------|---------------------------------------------------------------|
| (a) | $= h^\circ F(H^\circ \frac{H}{\exists} \text{est}(R)) h R^\circ$ | $S^\circ \frac{S}{\exists} \text{est}(R) \sqsubseteq R^\circ$ |
| (b) | $\sqsubseteq h^\circ F(R^\circ) h R^\circ$                       | $\text{MonoAlg}(h, R^\circ)$                                  |
| (c) | $\sqsubseteq R^\circ R^\circ$                                    | $R^\circ R^\circ \sqsubseteq R^\circ$                         |
| (d) | $\sqsubseteq R^\circ$                                            |                                                               |

$$\text{ThinCondition}(T, h, R, Q) \Rightarrow (\mu X : \frac{T^\circ}{3} \text{thin}(Q) P(F(X) h) \text{est}(R)) \sqsubseteq \frac{H}{3} \text{est}(R)$$

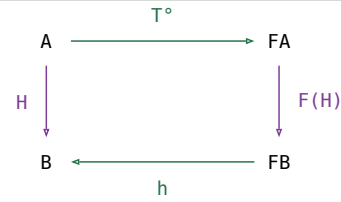
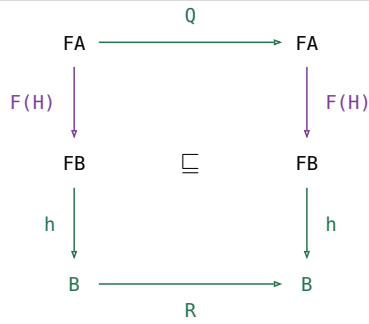
Theorem 9.2

an optimum over everything  $H$  returns is reached by taking the input apart every way  $T$  allows, dropping the parts that can never win, solving each of the rest and keeping one optimum

$$H \triangleq (T)^\circ (h) \triangleq H$$



|     |                                                                        |                                                                                        |
|-----|------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| (a) | $\frac{H}{3} \text{est}(R)$                                            | $QF(H) h \sqsubseteq F(H) h R, h \text{ monotonic on } R \text{ — (8.1a.6), (7.1.1a)}$ |
| (b) | $\sqsubseteq \frac{T^\circ}{3} \text{thin}(Q) P(F(X) h) \text{est}(R)$ |                                                                                        |



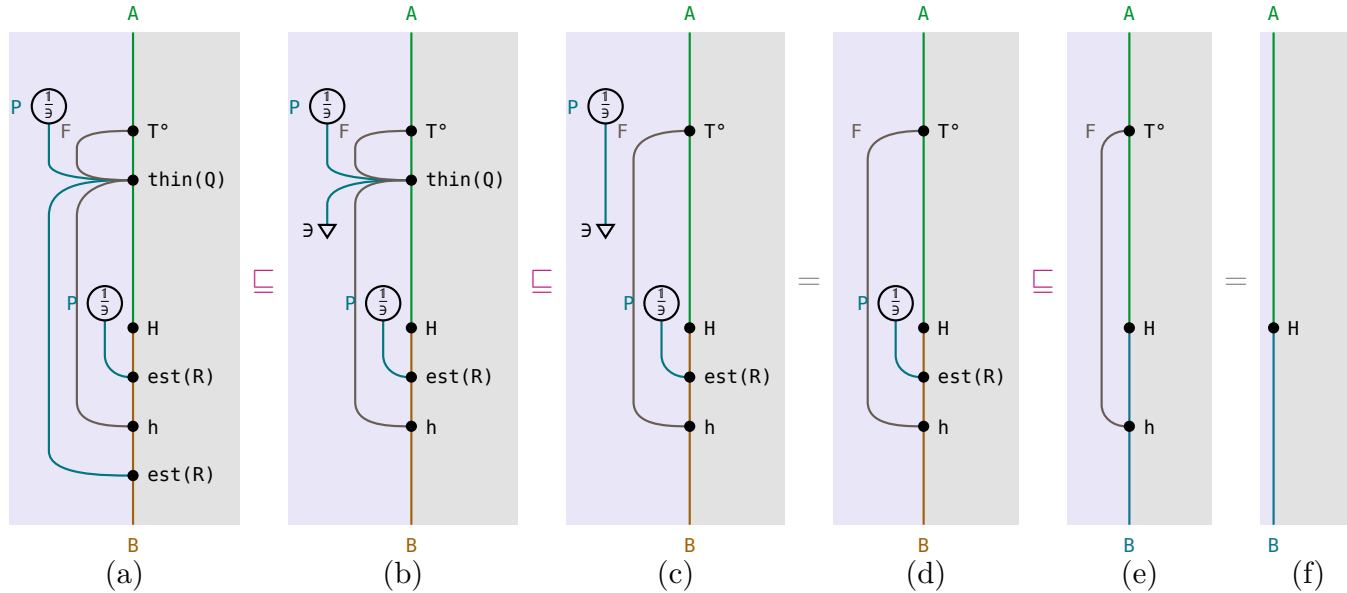
$T^\circ F(H) h = H$ : taking the input apart once by  $T^\circ$ , solving the parts by  $F(H)$  and assembling by  $h$  returns exactly what  $H$  returns

$QF(H) h \sqsubseteq F(H) h R$ : every result of thinning by  $Q$ , solving the parts by  $F(H)$  and assembling by  $h$  is reached by  $R$  from a result of solving by  $F(H)$  and assembling by  $h$  without  $Q$

$$T^\circ F(H) h = H \Rightarrow \frac{T^\circ}{3} \text{thin}(Q) P(F(\frac{H}{3} \text{est}(R)) h) \text{est}(R) \sqsubseteq H$$

(9.1.1e)

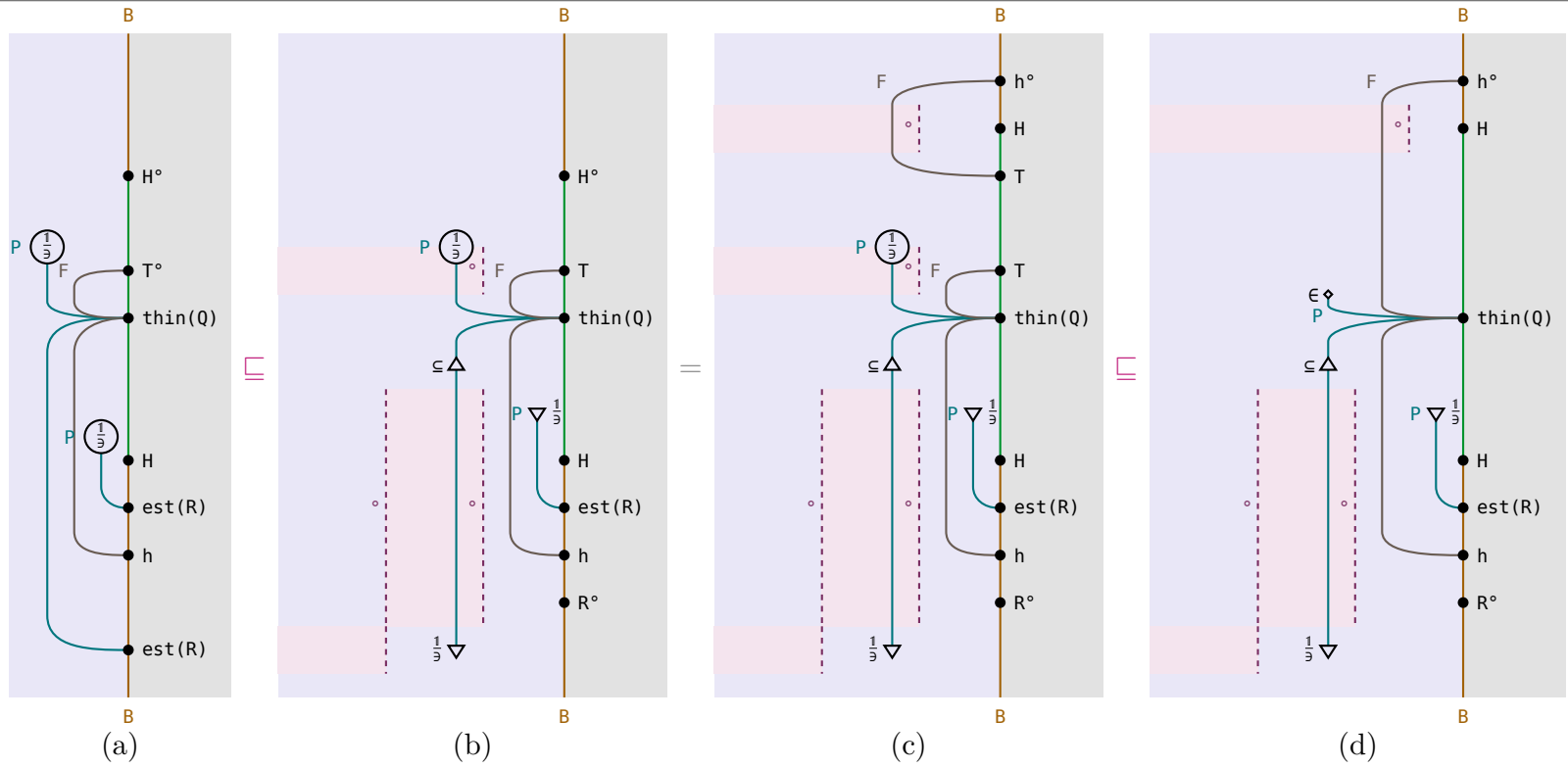
taking the input apart every way  $T$  allows ( $\frac{T^\circ}{3}$ ), dropping the parts  $\text{thin}(Q)$  rejects, solving each rest by  $M$  and keeping an optimum returns only what  $H$  returns



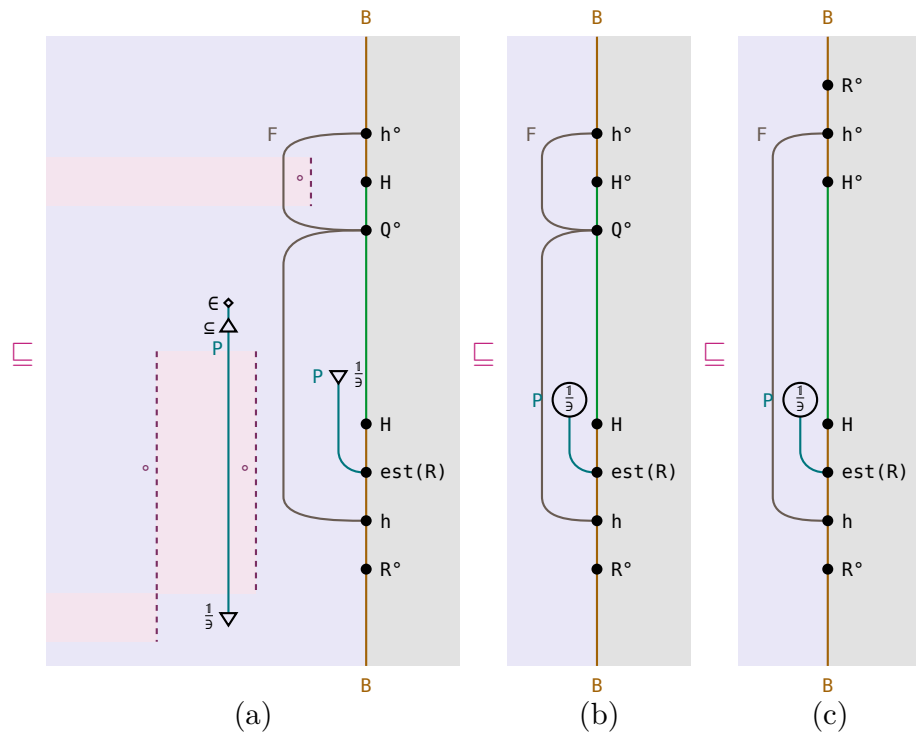
|     |                                                                                       |                                              |
|-----|---------------------------------------------------------------------------------------|----------------------------------------------|
| (a) | $\frac{T^\circ}{3} \text{thin}(Q) P(F(\frac{H}{3} \text{est}(R)) h) \text{est}(R)$    | $P(S) \text{est}(R) \sqsubseteq \exists S$   |
| (b) | $\sqsubseteq \frac{T^\circ}{3} \text{thin}(Q) \exists F(\frac{H}{3} \text{est}(R)) h$ | $\text{thin}(Q) \exists \sqsubseteq \exists$ |
| (c) | $\sqsubseteq \frac{T^\circ}{3} \exists F(\frac{H}{3} \text{est}(R)) h$                | $\frac{R}{3} \exists = R$                    |
| (d) | $= T^\circ F(\frac{H}{3} \text{est}(R)) h$                                            | $\frac{R}{3} \text{est}(R) \sqsubseteq S$    |
| (e) | $\sqsubseteq T^\circ F(H) h$                                                          | $T^\circ F(H) h = H$                         |
| (f) | $= H$                                                                                 |                                              |

$$T^\circ F(H) h = H \wedge Q F(H) h \subseteq F(H) h R \implies H^\circ \frac{T^\circ}{\exists} \text{thin}(Q) P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R) \subseteq R^\circ \quad (9.1.1f)$$

$H^\circ$  followed by the body at  $M$  is  $\mathbb{E}R^\circ$ : an answer of the body is never worse than an answer of  $H$  to the same input

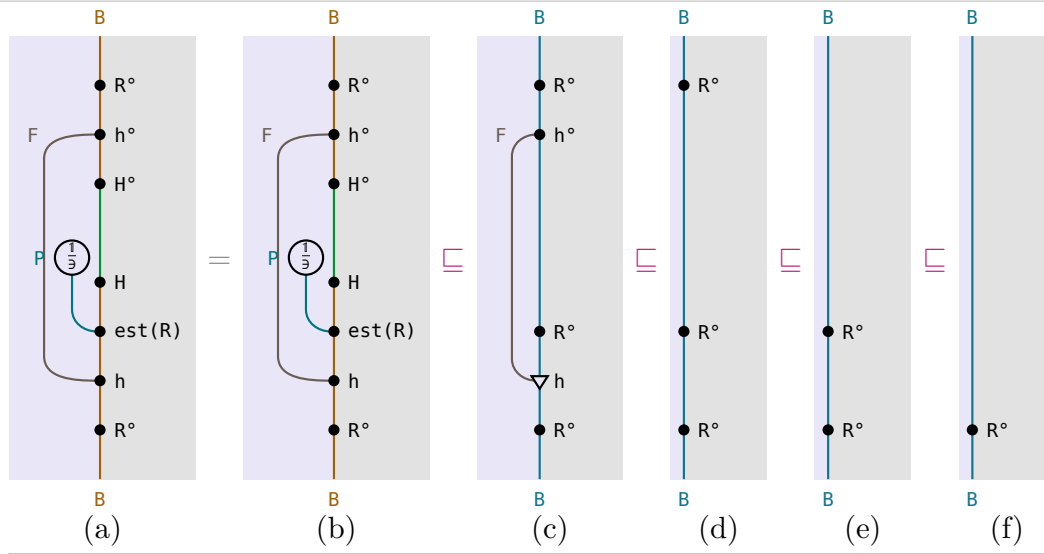


|     |                                                                                                                                       |                                                                 |
|-----|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| (a) | $H^\circ \frac{T^\circ}{\exists} \text{thin}(Q) P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R)$                                | $P(S) \text{est}(R) \subseteq \mathbb{E} \setminus (S R^\circ)$ |
| (b) | $\sqsubseteq H^\circ \frac{T^\circ}{\exists} \text{thin}(Q) (\mathbb{E} \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$    | $T^\circ F(H) h = H$                                            |
| (c) | $= h^\circ F(H^\circ) T \frac{T^\circ}{\exists} \text{thin}(Q) (\mathbb{E} \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$ | $\implies H^\circ = h^\circ F(H^\circ) T$                       |
| (d) | $\sqsubseteq h^\circ F(H^\circ) \mathbb{E} \text{thin}(Q) (\mathbb{E} \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$      | $T \frac{T^\circ}{\exists} \subseteq \mathbb{E}$                |
|     |                                                                                                                                       | $\mathbb{E} \text{thin}(Q) \subseteq Q^\circ \mathbb{E}$        |



|     |                                                                                                                           |                                                                            |
|-----|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| (a) | $\sqsubseteq h^\circ F(H^\circ) Q^\circ \mathbb{E} (\mathbb{E} \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$ | $S (S \setminus R) \subseteq R$                                            |
| (b) | $\sqsubseteq h^\circ F(H^\circ) Q^\circ F(\frac{H}{\exists} \text{est}(R)) h R^\circ$                                     | $Q F(H) h \subseteq F(H) h R$                                              |
| (c) | $\sqsubseteq R^\circ h^\circ F(H^\circ) F(\frac{H}{\exists} \text{est}(R)) h R^\circ$                                     | $\implies h^\circ F(H^\circ) Q^\circ \subseteq R^\circ h^\circ F(H^\circ)$ |

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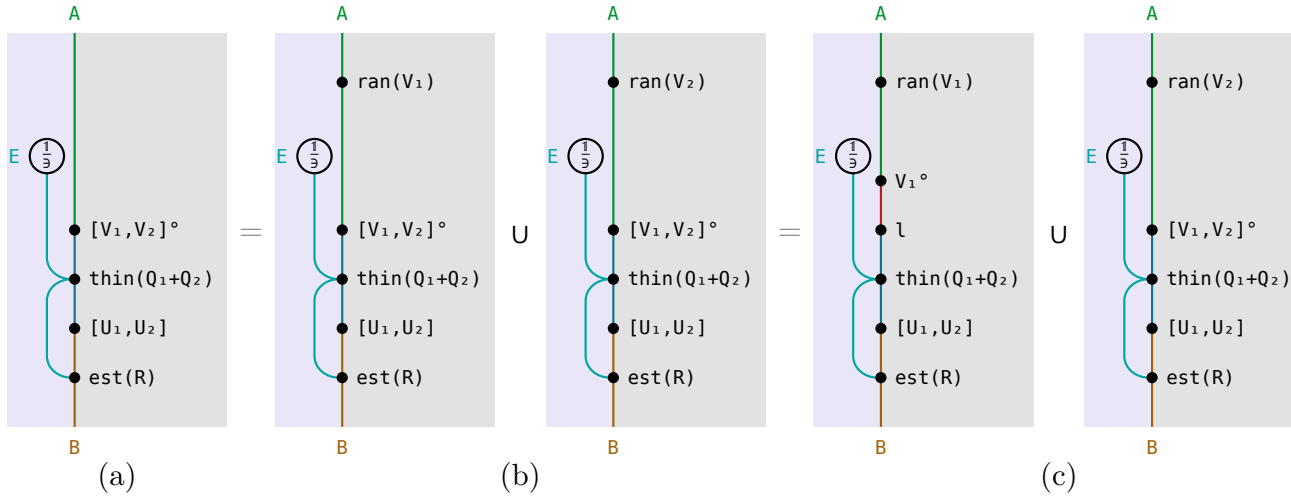


|       |                                                                                   |                                                               |
|-------|-----------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1 (a) | $R^\circ \ h^\circ \ F(H^\circ) \ F(\overset{H}{\exists} \ est(R)) \ h \ R^\circ$ | $t(f \ g) = t(f) \ t(g)$                                      |
| 2 (b) | $= R^\circ \ h^\circ \ F(H^\circ \ \overset{H}{\exists} \ est(R)) \ h \ R^\circ$  | $S^\circ \ \overset{S}{\exists} \ est(R) \sqsubseteq R^\circ$ |
| 3 (c) | $\sqsubseteq R^\circ \ h^\circ \ F(R^\circ) \ h \ R^\circ$                        | $\text{MonoAlg}(h, R^\circ)$                                  |
| 4 (d) | $\sqsubseteq R^\circ \ R^\circ \ R^\circ$                                         | $R^\circ \ R^\circ \sqsubseteq R^\circ$                       |
| 5 (e) | $\sqsubseteq R^\circ \ R^\circ$                                                   | $R^\circ \ R^\circ \sqsubseteq R^\circ$                       |
| 6 (f) | $\sqsubseteq R^\circ$                                                             |                                                               |

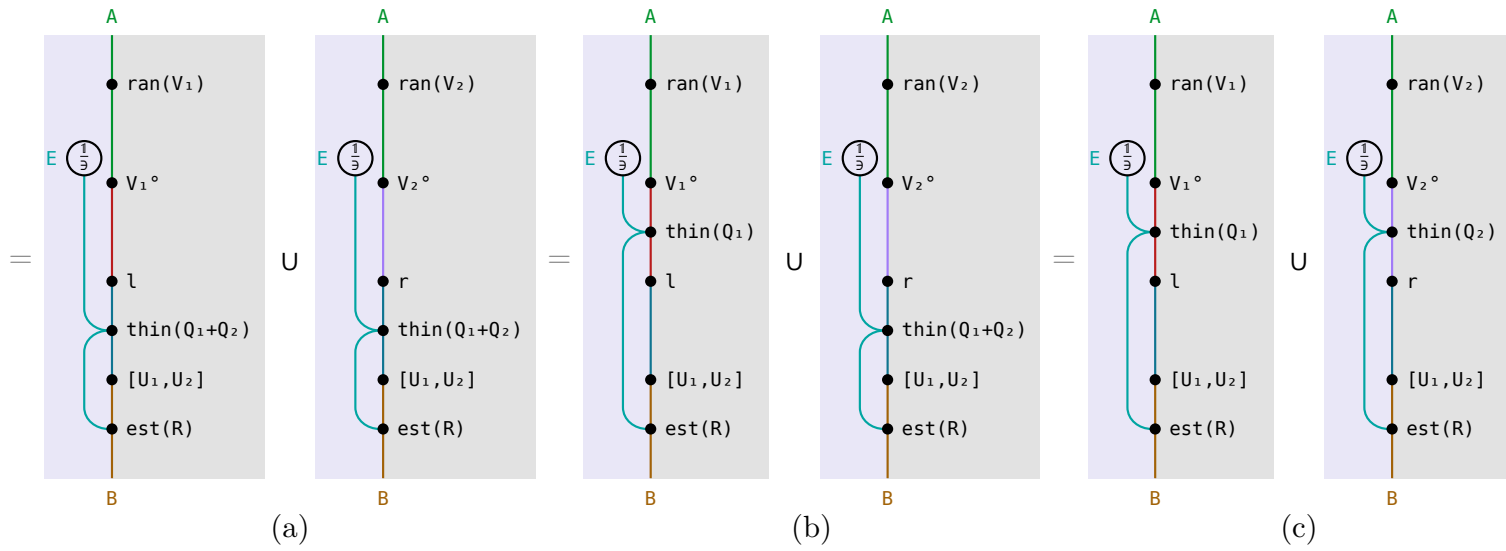
Proposition 9.1

$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R) = (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(U_1) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(U_2) \ \text{est}(R))$$

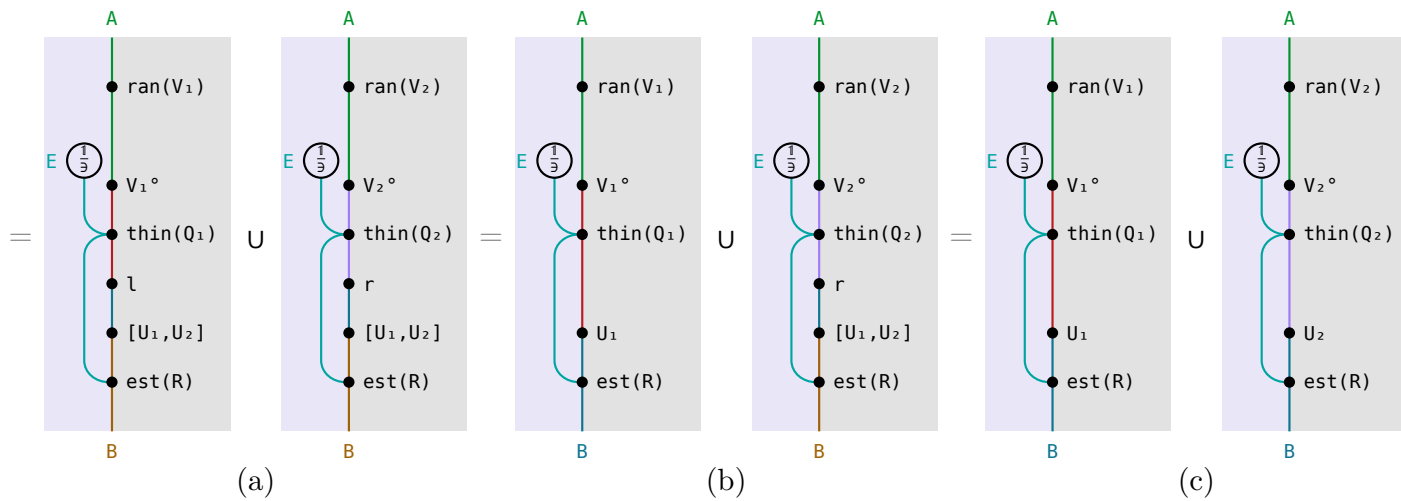
when  $V_1$  and  $V_2$  have disjoint ranges, the body over  $[V_1, V_2]$  runs the  $V_1$  problem on inputs  $V_1$  reaches and the  $V_2$  problem on inputs  $V_2$  reaches



|     |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                          |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)$                                                                                                                                        | $\frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R) = (\text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R))$ |
| (b) | $= (\text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R))$ | $V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{\exists} = \text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \ P(l)$                                                                                                                                                                           |
| (c) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R))$        | $V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} = \text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \ P(r)$                                                                                                                                                                           |



|     |                                                                                                                                                                                                                      |                                                           |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| (a) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R))$   | $P(l) \ \text{thin}(Q_1 + Q_2) = \text{thin}(Q_1) \ P(l)$ |
| (b) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(l) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(r) \ P([U_1, U_2]) \ \text{est}(R))$ | $P(r) \ \text{thin}(Q_1 + Q_2) = \text{thin}(Q_2) \ P(r)$ |
| (c) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(l) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(r) \ P([U_1, U_2]) \ \text{est}(R))$ | $P(R \ S) = P(R) \ P(S)$                                  |



|     |                                                                                                                                                                                                                      |                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| (a) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(l) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(r) \ P([U_1, U_2]) \ \text{est}(R))$ | $l \ [R, S] = R$ |
| (b) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(U_1) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(r) \ P([U_1, U_2]) \ \text{est}(R))$               | $r \ [R, S] = S$ |
| (c) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(U_1) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(U_2) \ \text{est}(R))$                             |                  |

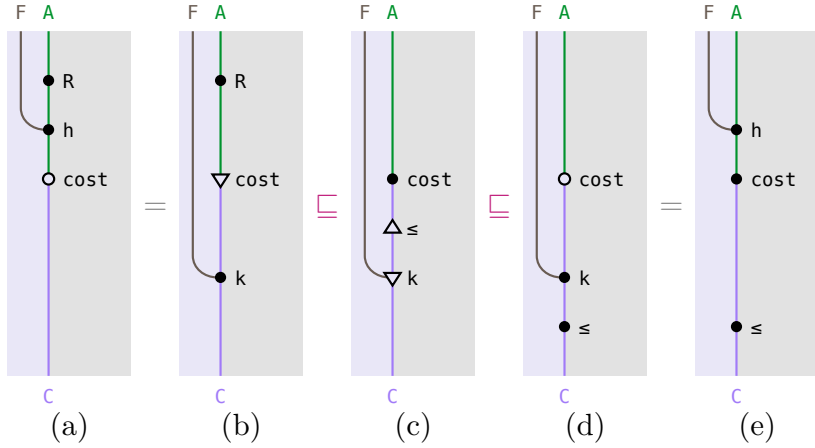
**Map(cost)  $\wedge$  R = cost  $\leq$  cost $^\circ$   $\wedge$  h cost = F(cost) k  $\wedge$  F( $\leq$ ) k  $\sqsubseteq$  k  $\leq \implies$  MonoAlg(h,R)** Proposition 9.2

if R compares two values by comparing their **costs** under  $\leq$ , and h then **cost** equals F(**cost**) then

a k monotonic on  $\leq$ , then h is monotonic on R

**Map(cost)  $\wedge$  R = cost  $\leq$  cost $^\circ$   $\wedge$  h cost = F(cost) k  $\wedge$  F( $\leq$ ) k  $\sqsubseteq$  k  $\leq \implies$  F(R) h cost  $\sqsubseteq$  h cost  $\leq$**

by the definition of R and shunting at the map **cost**, F(R)h  $\sqsubseteq$  hR is F(R)h cost  $\sqsubseteq$  h cost  $\leq$ , which the chain proves



|     |                                     |                                                                        |
|-----|-------------------------------------|------------------------------------------------------------------------|
| (a) | F(R) h cost                         | h cost = F(cost) k                                                     |
| (b) | = F(R) F(cost) k                    | Map(f) $\wedge$ R = f S f $^\circ$<br>$\implies$ R f $\sqsubseteq$ f S |
| (c) | $\sqsubseteq$ F(cost) F( $\leq$ ) k | F( $\leq$ ) k $\sqsubseteq$ k $\leq$                                   |
| (d) | $\sqsubseteq$ F(cost) k $\leq$      | h cost = F(cost) k                                                     |
| (e) | = h cost $\leq$                     |                                                                        |

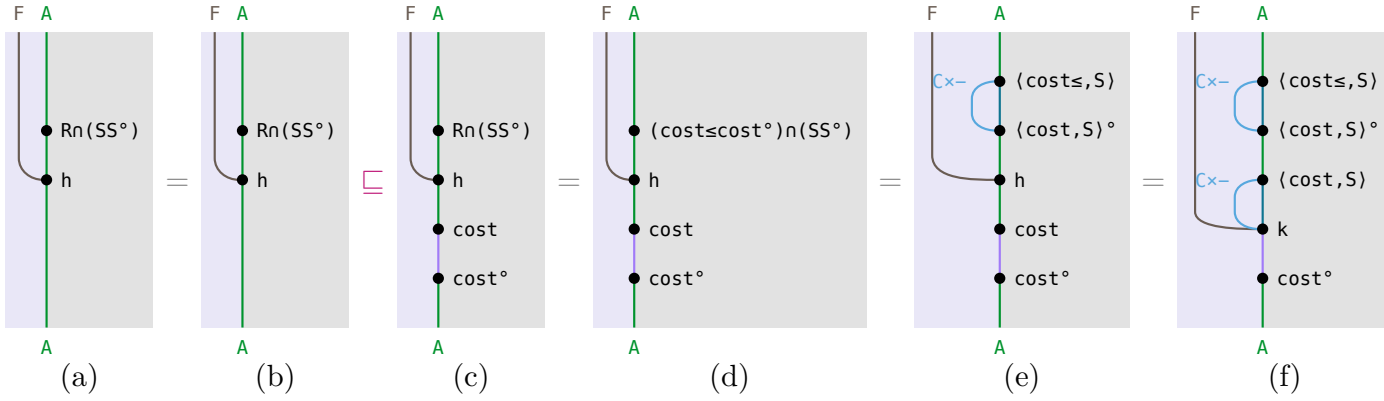
(9.1.1h)

|   | the condition                                                                                                                                                                                                | how it is discharged                                                               |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1 | F(R)h $\sqsubseteq$ hR<br>Proposition 9.2, R $\triangleq$ cost $\leq$ cost $^\circ$ , h cost = F(cost)k, F( $\leq$ ) k $\sqsubseteq$ k $\leq$                                                                | monotonicity when the cost is itself a fold with a step k<br>monotonic on $\leq$   |
| 2 | F(Rn(H $^\circ$ H))h $\sqsubseteq$ hR<br>Proposition 9.3, R $\triangleq$ cost $\leq$ cost $^\circ$ , h cost = F((cost, H $^\circ$ ))k,<br>F( $\leq \times 1$ ) k $\sqsubseteq$ k $\leq$ , H $^\circ$ simple; | monotonicity <b>in context</b> : k may also read the input the part was built from |
| 3 | QF(H)h $\sqsubseteq$ F(H)hR at Q $\triangleq$ F(U,V)<br>Proposition 9.4, U, V preorders, F(U,R)h $\sqsubseteq$ hR, Vh $\sqsubseteq$ hR                                                                       | both conditions at once, split along the two arguments of a bifunctor              |

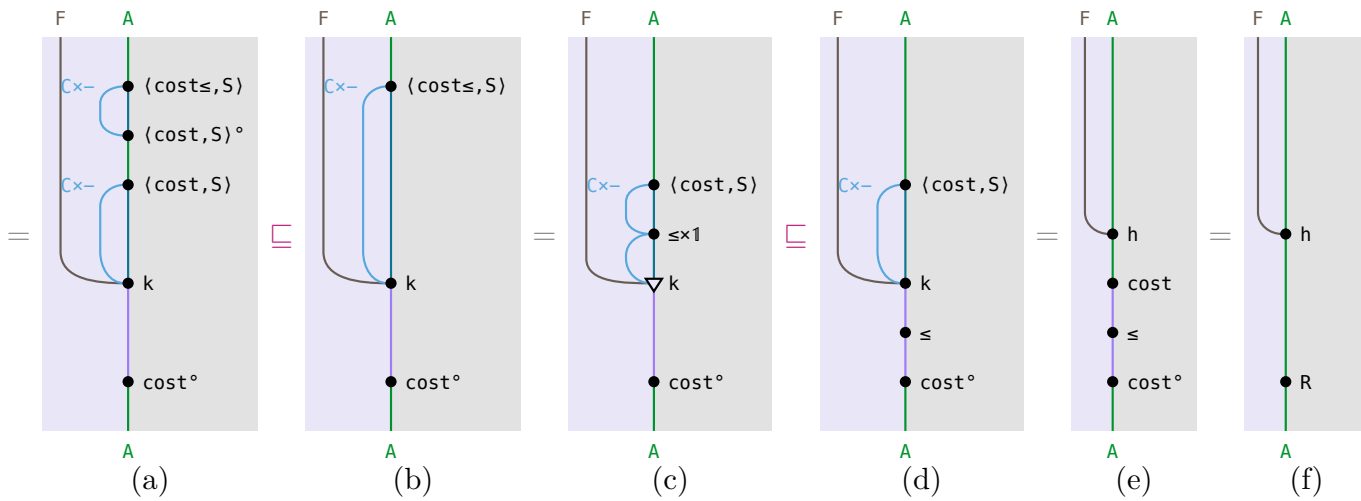
**Map(cost)  $\wedge$  Simple(S)  $\wedge$  R = cost  $\leq$  cost $^\circ$   $\wedge$  h cost = F((cost,S)) k  $\wedge$  F( $\leq \times 1$ ) k  $\sqsubseteq$  Proposition 9.3**

**k  $\leq \Rightarrow$  F(R  $\cap$  (S S $^\circ$ )) h  $\sqsubseteq$  h R**

improving each part by R within its S-context, then assembling by h, is below hR



|     |                                                                               |                                                                                                              |
|-----|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| (a) | F(R $\cap$ (S S $^\circ$ )) h                                                 |                                                                                                              |
| (b) | = F(R $\cap$ (S S $^\circ$ )) h                                               | Map(f)<br>$\Rightarrow 1 \sqsubseteq f f^\circ$                                                              |
| (c) | $\sqsubseteq$ F(R $\cap$ (S S $^\circ$ )) h cost cost $^\circ$                | R = cost $\leq$ cost $^\circ$                                                                                |
| (d) | = F((cost $\leq$ cost $^\circ$ ) $\cap$ (S S $^\circ$ )) h cost cost $^\circ$ | $\langle X, Y \rangle \langle R, S \rangle^\circ = \langle X R^\circ \rangle \cap \langle Y S^\circ \rangle$ |
| (e) | = F((cost $\leq$ , S) (cost, S) $^\circ$ ) h cost cost $^\circ$               | h cost = F((cost, S)) k                                                                                      |
| (f) | = F((cost $\leq$ , S) (cost, S) $^\circ$ ) F((cost, S)) k cost $^\circ$       | t(f g) = t(f) t(g)                                                                                           |



|     |                                                                     |                                                                                                  |
|-----|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| (a) | = F((cost $\leq$ , S) (cost, S) $^\circ$ (cost, S)) k cost $^\circ$ |                                                                                                  |
| (b) | $\sqsubseteq$ F((cost $\leq$ , S)) k cost $^\circ$                  | $\langle X, Y \rangle \langle R \times 1 \rangle = \langle X R, Y \rangle$<br>t(f g) = t(f) t(g) |
| (c) | = F((cost, S)) F( $\leq \times 1$ ) k cost $^\circ$                 | F( $\leq \times 1$ ) k $\sqsubseteq$ k $\leq$                                                    |
| (d) | $\sqsubseteq$ F((cost, S)) k $\leq$ cost $^\circ$                   | h cost = F((cost, S)) k                                                                          |
| (e) | = h cost $\leq$ cost $^\circ$                                       | R = cost $\leq$ cost $^\circ$                                                                    |
| (f) | = h R                                                               |                                                                                                  |

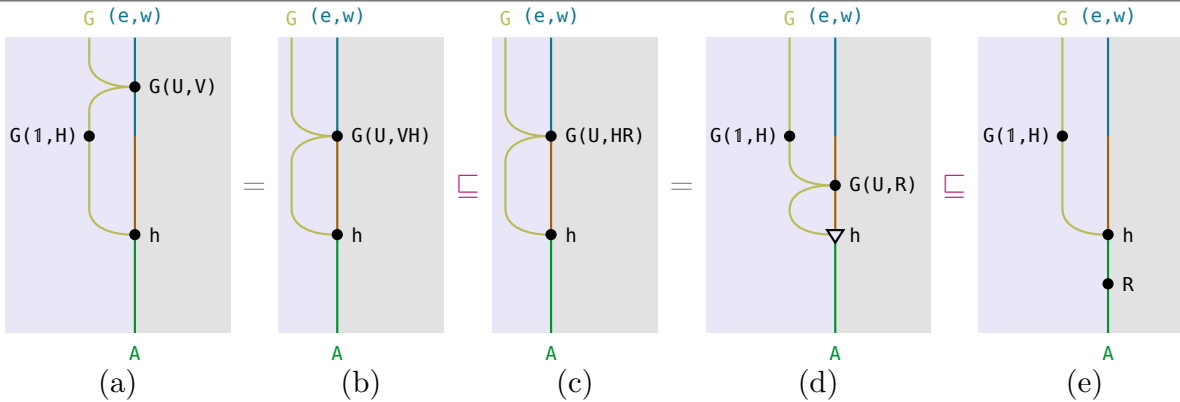
**G(U,R) h  $\sqsubseteq$  h R  $\wedge$  V H  $\sqsubseteq$  H R  $\Rightarrow$  G(U,V) G(1,H) h  $\sqsubseteq$  G(1,H) h R**

Proposition 9.4

thinning the parts by U in the first argument and by V in the second (G(U,V)), then solving by H

and assembling by h, gives only what solving and assembling and then improving by R gives

(G(1,H)hR)



|     |                           |                                   |
|-----|---------------------------|-----------------------------------|
| (a) | G(U,V) G(1,H) h           | F(R R', S S') = F(R, S) F(R', S') |
| (b) | = G(U, V H) h             | V H $\sqsubseteq$ H R             |
| (c) | $\sqsubseteq$ G(U, H R) h | F(R R', S S') = F(R, S) F(R', S') |
| (d) | = G(1,H) G(U,R) h         | G(U,R) h $\sqsubseteq$ h R        |
| (e) | $\sqsubseteq$ G(1,H) h R  |                                   |

## §9.2. The string edit problem

- $(\exists a. o = \text{cpy}(a)) \vee (\exists a. o = \text{del}(a)) \vee (\exists a. o = \text{ins}(a))$   
an edit operation copies, deletes or inserts one character
- $\text{FX} = 1 + (0p \times X)$   
 $F(0p, X)$  is the empty sequence or one operation paired with an  $X$
- $\text{edit} = ([\text{base}, \text{step}])$   
 $\text{edit}$  is the fold of  $[\text{base}, \text{step}]$  over an edit sequence, returning the two strings it reconstitutes
- $\text{length} = ([\text{zero}, \pi_2 \text{ succ}])$   
the fold of  $[\text{zero}, \pi_2 \text{ succ}]$  counts the operations of a sequence
- $R = \text{length} \leq \text{length}^\circ$   
 $R$  relates  $es$  to  $fs$  when  $es$  is no longer than  $fs$
- $V = \text{suffix}^\circ \times \text{suffix}^\circ$   
 $V$  relates two pairs of strings when each string of the first is a suffix of the matching string of the second
- $Q(\text{inl}(u), \text{inl}(u))$   
 $Q$  is the identity on the **base** summand
- $Q(\text{inr}(p), \text{inr}(q)) \iff V(\pi_2(p), \pi_2(q))$   
on the **step** summand  $Q$  compares any two operations ( $U \triangle T$ ) and the two pairs of strings by  $V$
- $\text{empty}(p, q) \iff p = q \wedge p = ([], [])$   
 $\text{empty}$  is the coreflexive holding only at the pair of two empty strings
- $q \in \text{unstep}(p) \implies \text{step}(\text{inr}(q)) = p$   
every decomposition  $\text{unstep}$  returns is sent back to its input by  $\text{step}$

(9.2a)

| <b>x</b>                | <b><math>\alpha(x)</math></b> |
|-------------------------|-------------------------------|
| 1 $\text{inl}(u)$       | $\text{wrap}(u)$              |
| 2 $\text{inr}((o, es))$ | $[o]\#es$                     |

(9.2b)

| <b>x</b>                                  | <b><math>[\text{base}, \text{step}](x)</math></b> |
|-------------------------------------------|---------------------------------------------------|
| 1 $\text{inl}(u)$                         | $([], [])$                                        |
| 2 $\text{inr}((\text{cpy}(a), (xs, ys)))$ | $([a]\#xs, [a]\#ys)$                              |
| 3 $\text{inr}((\text{del}(a), (xs, ys)))$ | $([a]\#xs, ys)$                                   |
| 4 $\text{inr}((\text{ins}(a), (xs, ys)))$ | $(xs, [a]\#ys)$                                   |

(9.2c)

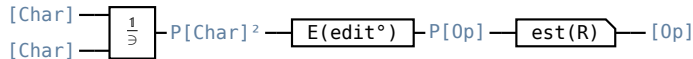
| <b>p</b>               | <b><math>\text{unstep}(p)</math></b>                                                                   |
|------------------------|--------------------------------------------------------------------------------------------------------|
| 1 $([], [])$           | $[]$                                                                                                   |
| 2 $([a]\#xs, [])$      | $[(\text{del}(a), (xs, []))]$                                                                          |
| 3 $([], [b]\#ys)$      | $[(\text{ins}(b), ([], ys))]$                                                                          |
| 4 $([a]\#xs, [b]\#ys)$ | if $a = b$ then $[(\text{cpy}, xs, ys)]$ else $[(\text{del}, xs, [b]\#ys), (\text{ins}, [a]\#xs, ys)]$ |

(9.2d)

$$\begin{aligned} \text{mle} = & (\text{empty} \rightarrow \text{nil}, \text{unstep } \text{list}((1 \times \text{mle}) \text{cons}) \text{minlist}(\text{R})), (\mu X : \frac{[\text{base}, \text{step}]^\circ}{3} \text{thin}(\text{Q}) \text{P}(\text{F}(\text{X}) \\ & [\text{nil}, \text{cons}]) \text{est}(\text{R})) \sqsubseteq \frac{\text{edit}^\circ}{3} \text{est}(\text{R}) \end{aligned} \quad (9.2e)$$

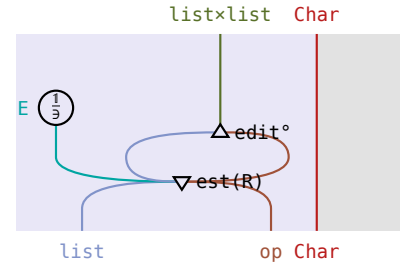
a shortest edit sequence from which both strings can be reconstituted is one pass over the two of them, each step copying, deleting or inserting one character and the best sequence for what is left taken from the entries already computed

## circuit



the specification —  $\triangleq$  edit

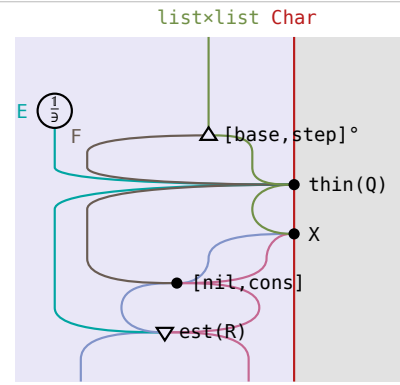
## Hinze–Marsden



list op Char



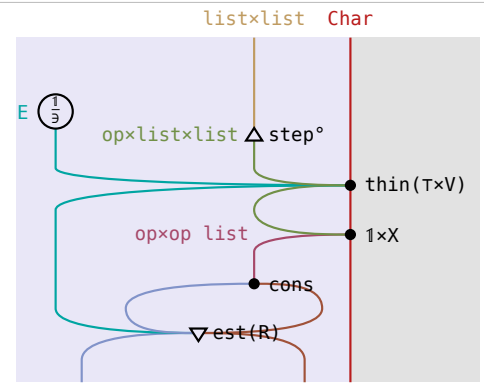
at  $Q \triangleq 1 + (U \times V)$ . Monotonicity  $F(R) \alpha \sqsubseteq \alpha R$  is Proposition 9.2 at  $\text{length} \triangleq ([\text{zero}, \pi_2 \text{succ}])$  with  $\text{succ}$  monotonic on  $\leq$ , so  $\text{cons}$  is monotonic on  $R$ . The thinning condition  $\sqsubseteq QF(1, \text{edit}^\circ) \alpha \sqsubseteq F(1, \text{edit}^\circ) \alpha R$  over  $F(0p, [\text{Char}] \times [\text{Char}])$  is Proposition 9.4 at  $U \triangleq T$  —  $F(T, R) \alpha \sqsubseteq \alpha R$ , left as an exercise, so any two operations may be compared — and  $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$ , —  $\text{edit}(1 \times \text{suffix}) \sqsubseteq R^\circ \text{edit}, \text{edit}(\text{suffix} \times 1) \sqsubseteq R^\circ \text{edit}$ .  $\text{suffix} = \text{tail}^*$  and  $BA \sqsubseteq CB \Rightarrow BA^* \sqsubseteq C^*B$  cut those to one step each: drop the operation that produced the head, or weaken its  $\text{cpy}$  to a  $\text{del}$ , never lengthening the sequence



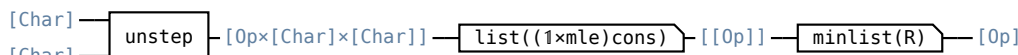
list op Char



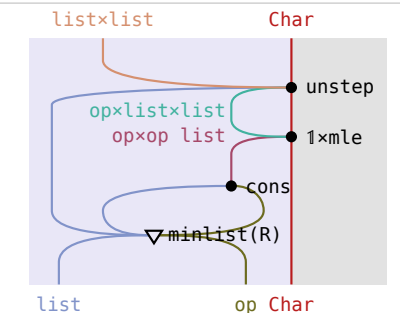
= Proposition 9.1: **base** and **step** have disjoint ranges, so the fixed point splits into one branch per summand — **empty**, the coreflexive on  $(xs, ys)$  with both lists empty, is where **base** returns



list op Char



$\text{unstep}$  implements  $\frac{\text{step}^\circ}{3} \text{thin}(U \times V)$  — at most two decompositions survive, a copy beating a delete or an insert wherever it is available — and  $\text{minlist}(R)$  implements  $\text{est}(R)$ . The same subproblem is solved many times over, so the running time is exponential in the two lengths

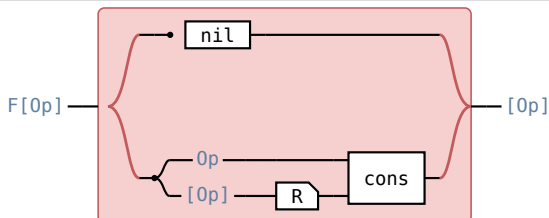


list op Char

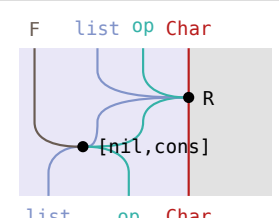
$$F(R) [\text{nil}, \text{cons}] \sqsubseteq [\text{nil}, \text{cons}] R \quad (9.2f)$$

comparing the tails by  $R$  and then putting an operation in front ( $F(R) \alpha$ ) relates only sequences that  $R$  also relates once the operation is in front ( $\alpha R$ )

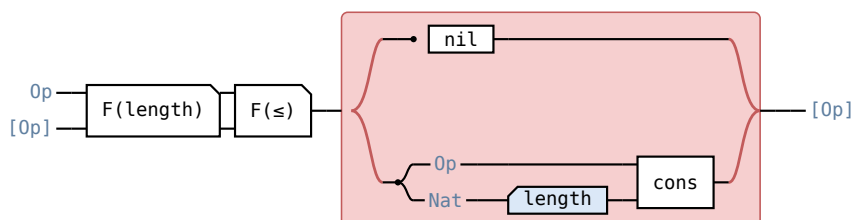
## circuit



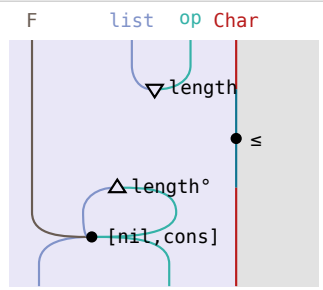
## Hinze–Marsden



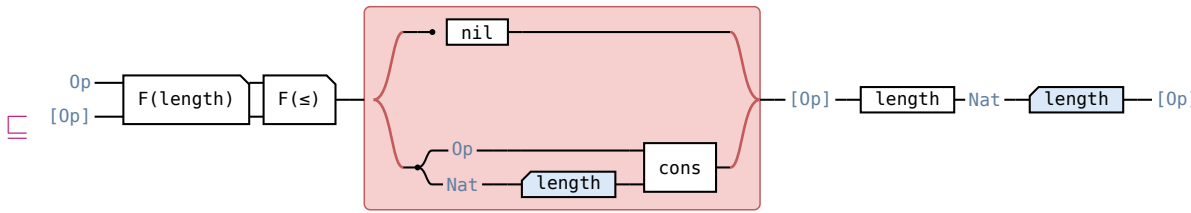
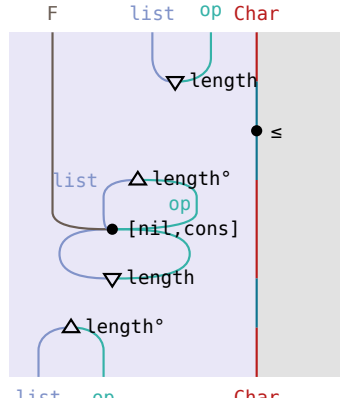
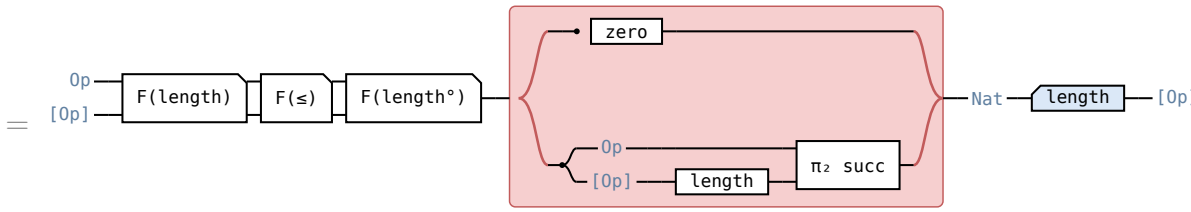
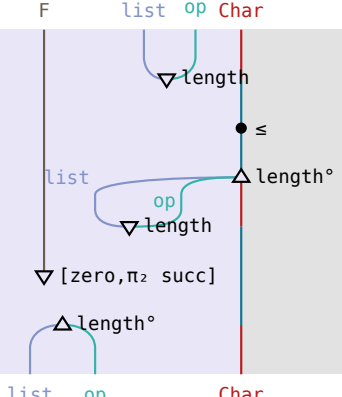
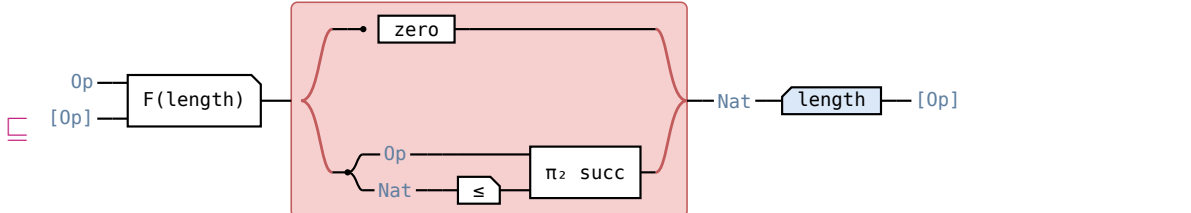
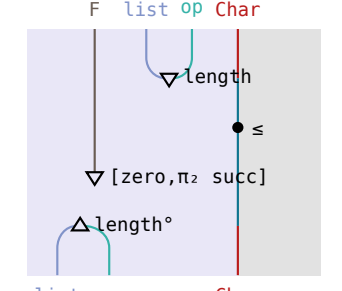
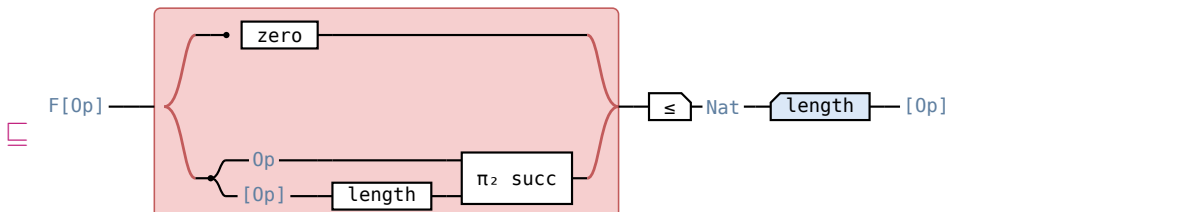
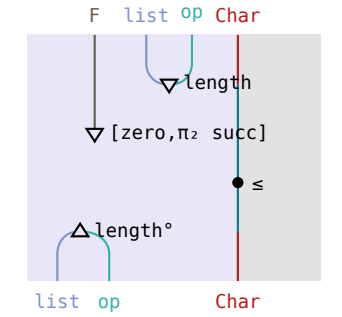
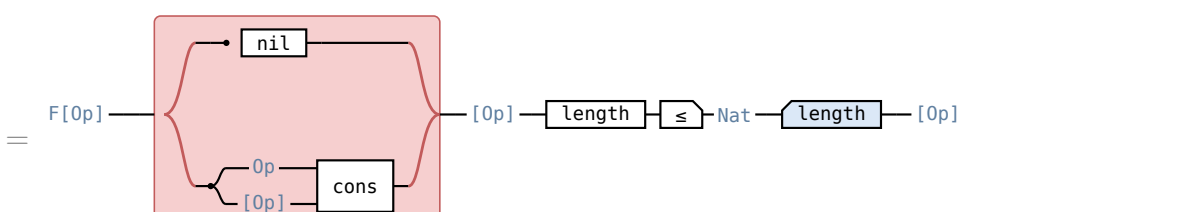
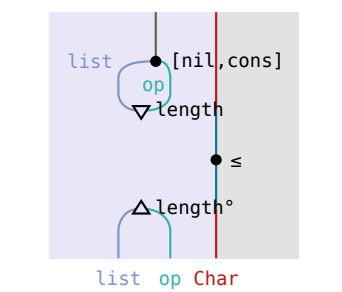
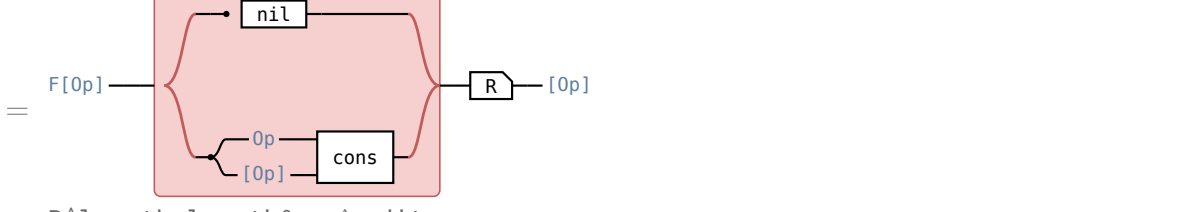
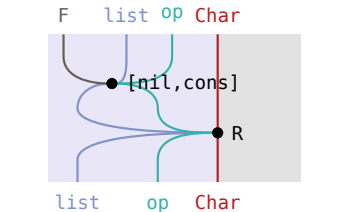
list op Char



$R \triangleq \text{length} \leq \text{length}^\circ$  —  $\triangleq$  edit;  $F$  preserves composition

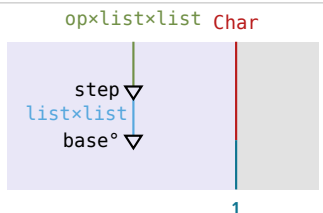
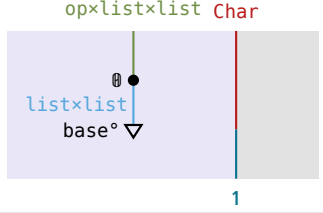
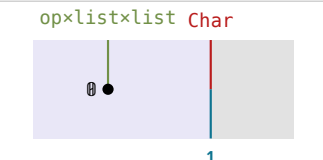


list op Char

| circuit                                                                                                                                                                                                                                                              | Hinze–Marsden                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  <p>length is a map, so entire: <math>1 \sqsubseteq \text{length} \text{ length}^\circ</math></p>                                                                                  |    |
|  <p><math>\alpha \text{ length} = F(\text{length})[\text{zero}, \pi_2 \text{ succ}] : \text{length} \triangle ([\text{zero}, \pi_2 \text{ succ}]) \triangle \text{edit}</math></p> |    |
|  <p>length is a map, so simple, and F is monotonic: <math>F(\text{length}^\circ)F(\text{length}) \sqsubseteq F(1) = 1</math></p>                                                 |  |
|  <p>SUCC is monotonic on <math>\leq</math>: <math>F(\leq)[\text{zero}, \pi_2 \text{ succ}] \sqsubseteq [\text{zero}, \pi_2 \text{ succ}] \leq</math></p>                         |  |
|  <p><math>F(\text{length})[\text{zero}, \pi_2 \text{ succ}] = \alpha \text{ length}</math> again</p>                                                                             |  |
|  <p><math>R \triangle \text{length} \leq \text{length}^\circ \triangle \text{edit}</math></p>                                                                                    |  |



| $\text{edit } V^\circ \sqsubseteq R^\circ \text{ edit}$ (9.2h)<br>cutting a front off either string an edit sequence produces $(\text{edit } V^\circ)$ leaves a pair that a sequence no longer produces $(R^\circ \text{ edit})$                                                                                                                                                         |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                  | Hinze–Marsden |
|                                                                                                                                                                                                                                                                                                                                                                                          |               |
| $=$ <p><math>V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ</math>, and <math>\times</math> preserves composition</p>                                                                                                                                                                                                                                                       |               |
| $\sqsubseteq$ <p><math>\text{edit } (\text{suffix} \times 1) \sqsubseteq R^\circ \text{ edit}</math>: drop the operation that produced the head, or weaken its <b>cpy</b> to an <b>ins</b>. Proved by induction at <b>suffix</b> directly, not through <math>\text{suffix} = \text{tail}^*</math> and <math>BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B</math> as the book does</p> |               |
| $\sqsubseteq$ <p><math>\text{edit } (1 \times \text{suffix}) \sqsubseteq R^\circ \text{ edit}</math>, the mirror image: <b>cpy</b> weakens to a <b>del</b></p>                                                                                                                                                                                                                           |               |
| $\sqsubseteq$ <p><math>R^\circ</math> is transitive, <math>\leq</math> being so</p>                                                                                                                                                                                                                                                                                                      |               |

| $\text{step } \text{base}^\circ = \emptyset$ <span style="float: right;">(9.2i)</span><br>no pair of strings is both a result of <b>step</b> and the result $([], [])$ of <b>base</b>                                                  |                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| circuit                                                                                                                                                                                                                                | Hinze–Marsden                                                                        |
|                                                                                                                                                       |   |
|  <p><math>\text{base}</math> returns only <math>([], [])</math>, so <math>\text{base}^\circ = \text{empty } \text{base}^\circ</math></p>              |   |
|  <p><math>\text{step } \text{empty} = \emptyset</math>: <b>cpy</b> and <b>del</b> put a character on the left string, <b>ins</b> one on the right</p> |   |
|  <p><math>\emptyset</math> composed with anything is <math>\emptyset</math></p>                                                                      |  |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\text{mle}(xs, ys) = \text{head}(\text{column}(xs, ys))$ , $\text{column}(xs, ys) = [\text{mle}(u, ys) \mid u \leftarrow \text{tails}(xs)]$<br>$= \text{column}(xs) = ([\text{fstcol}(xs), \text{nextcol}(xs)])$ , $\text{fstcol} = \text{list}(\text{del}) \text{ tails}$<br>the tabulation: $\text{mle}(xs, ys)$ needs $\text{mle}(u, v)$ for every tail $u$ of $xs$ and $v$ of $ys$ , so the columns are built right to left                          |
| 2 | $\text{column}(xs, [b] \# ys) = \text{nextcol}(xs, (b, \text{column}(xs, ys)))$<br>$= \text{nextcol}(xs)(b, us) = ([\text{base}(b, \text{last}(us)), \text{step}(b)])(xus)$ , $xus = \text{zip}(xs, \text{zip}(\text{init}(us), \text{tail}(us)))$<br>each column is a fold built bottom to top, over $xs$ zipped with the adjacent pairs of the column to its right                                                                                      |
| 3 | $\text{base}(b, u) = [[\text{ins}(b)] \# u]$<br>$= \text{step}(b)((a, (u, v)), ws) = (a \rightarrow b \rightarrow [[\text{cpy}(a)] \# v] \# ws, [\text{bmin}(R)([\text{del}(a)] \# w, [\text{ins}(b)] \# u)] \# ws)$<br>$w = \text{head}(ws)$ ; these <b>base</b> , <b>step</b> are not <b>edit</b> 's. An entry depends on the one below it (a delete), the one to its right (an insert), and the one below that (a copy) — quadratic in the two lengths |

### §9.3. Optimal bracketing

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <p>tree <math>A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)</math>, <math>FX = A + X^2</math>, so <math>F(R) = 1 + R^2</math>; <math>h \triangleq [\text{tip}, \text{bin}]</math>, <math>\text{flatten} \triangleq ([\text{wrap}, \text{cat}]) : \text{tree } A \rightarrow \text{list}^+ A</math>, <math>H = \text{flatten}^\circ</math>.</p> <p><math>\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}])</math>, <math>\text{opt} \triangleq (\text{zero}, \text{st})</math>, <math>\text{opb } ((cx, sx), (cy, sy)) = (\text{cb } (sx, sy) + cx + cy, \text{sb } (sx, sy))</math>.</p> <p><b>sb</b> associative, so <math>\text{size} = \text{flatten } \text{sz}</math> for a map <math>\text{sz}</math> <math>R \triangleq \text{cost} \leq \text{cost}^\circ</math>, <math>g \triangleq [\text{zero}, (1 \times \text{sz})^2 \text{ opb } \pi_1]</math>, single the coreflexive on singleton lists.</p> <p><math>\text{splits} \triangleq (\text{inits}^+, \text{tails}^+) \text{ zip}</math>, an implementation of <math>\frac{\text{cat}^\circ}{\text{sz}}</math>; <math>\text{array} \triangleq \text{inits } \text{list}(\text{row})</math>, <math>\text{row} \triangleq \text{tails } \text{list}(\text{mct})</math>, <math>\text{col} \triangleq \text{inits } \text{list}(\text{mct})</math>.</p> <p><math>\text{mix} \triangleq \text{zip } \text{list}(\text{bin}) \text{ minlist}(R)</math>, <math>\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc}</math>, <math>\text{process} \triangleq ((\text{tip } \text{wrap}) \times 1) \text{ loop}(\text{next})</math>.</p> | (9.3a) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|

| $\text{Assoc}(\text{sb}) \Rightarrow F(\langle \text{cost}, \text{flatten} \rangle) \text{ g} = [\text{tip}, \text{bin}] \text{ cost}$ (9.5)<br>pairing each subtree with its cost and its flattening and then applying <b>g</b> gives what building the tree by <b>[tip, bin]</b><br>and taking its <b>cost</b> gives                                                                                                                                                                                                                                                                                                                          |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Hinze–Marsden |
| $F(\text{tree}(A)) \rightarrow \boxed{F(\langle \text{cost}, \text{flatten} \rangle)} \rightarrow F(\text{Int} \times \text{list}^+(A)) \rightarrow \boxed{\text{g}} \rightarrow \text{Int}$                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \text{l} \rightarrow A \rightarrow \text{zero} \\ \text{r} \rightarrow \text{tree}(A) \rightarrow \left\{ \begin{array}{l} \text{cost} \\ \text{flatten} \rightarrow \text{list}^+(A) \rightarrow \text{sz} \end{array} \right\} \\ \text{r} \rightarrow \text{tree}(A) \rightarrow \left\{ \begin{array}{l} \text{cost} \\ \text{flatten} \rightarrow \text{list}^+(A) \rightarrow \text{sz} \end{array} \right\} \end{array} \right\} \rightarrow \text{opb} \rightarrow \text{Int}$ <p>definition of <b>g</b>; coproducts and products <math>\triangleq \text{flatten}</math></p> |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \text{l} \rightarrow A \rightarrow \text{zero} \\ \text{r} \rightarrow \text{tree}(A) \rightarrow \left\{ \begin{array}{l} \text{cost} \\ \text{size} \end{array} \right\} \\ \text{r} \rightarrow \text{tree}(A) \rightarrow \left\{ \begin{array}{l} \text{cost} \\ \text{size} \end{array} \right\} \end{array} \right\} \rightarrow \text{opb} \rightarrow \text{Int}$ <p><math>\text{flatten sz} = \text{size}</math>, since <b>sb</b> is associative <math>\triangleq \text{flatten}</math></p>                                                                                |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \text{l} \rightarrow A \rightarrow \text{zero} \\ \text{r} \rightarrow \text{tree}(A) \rightarrow \text{bin} \rightarrow \text{tree}(A) \rightarrow \left\{ \begin{array}{l} \text{cost} \rightarrow \text{Int} \\ \text{size} \rightarrow S \end{array} \right\} \end{array} \right\} \rightarrow \text{Int}$ <p><math>\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}])</math>, at a node <math>\text{bin}(\text{cost}, \text{size}) = \langle \text{cost}, \text{size} \rangle^2 \text{ opb}</math></p>                                              |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \text{l} \rightarrow A \rightarrow \text{tip} \rightarrow \text{tree}(A) \rightarrow \text{cost} \\ \text{r} \rightarrow \text{tree}(A) \rightarrow \text{bin} \rightarrow \text{tree}(A) \rightarrow \text{cost} \end{array} \right\} \rightarrow \text{Int}$ <p><math>\text{tip cost} = \text{zero}</math>; products, <math>\langle \text{cost}, \text{size} \rangle \pi_1 = \text{cost}</math></p>                                                                                                                                                                                |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \text{l} \rightarrow A \rightarrow \text{tip} \\ \text{r} \rightarrow \text{tree}(A) \rightarrow \text{bin} \end{array} \right\} \rightarrow \text{tree}(A) \rightarrow \text{cost} \rightarrow \text{Int}$ <p>coproducts, <math>[\text{tip cost}, \text{bin cost}] = [\text{tip}, \text{bin}] \text{ cost}</math></p>                                                                                                                                                                                                                                                               |               |

| $F(\leq \times 1) \ g \sqsubseteq g \leq$ <span style="float: right;">(9.6)</span>                                                                                                                         |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| raising the cost of either subtree ( $\leq \times 1$ in both slots) and then applying $g$ returns a value at least what $g$ returns on the costs before raising ( $g \leq$ )                               |               |
| circuit                                                                                                                                                                                                    | Hinze–Marsden |
| $F(\text{Int} \times \text{list}^+(A)) \rightarrow \boxed{F(\leq \times 1)} \rightarrow \boxed{g} \rightarrow \text{Int}$                                                                                  |               |
| $A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$<br>$=$<br>definition of $g \triangleq \text{flatten}$                                            |               |
| $A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$<br>$\sqsubseteq$<br>definition of $\text{opb}$ , and $+$ monotonic                               |               |
| $A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$<br>$\sqsubseteq$<br>$\leq$ reflexive, so $\text{zero} \sqsubseteq \text{zero} \leq$ ; coproducts |               |
| $F(\text{Int} \times \text{list}^+(A)) \rightarrow \boxed{g} \rightarrow \boxed{\leq} \rightarrow \text{Int}$<br>$=$ definition of $g \triangleq \text{flatten}$                                           |               |

$$\text{mct} = (\text{single} \rightarrow \text{head tip}, (\text{init col}, \text{tail row}) \text{ mix}), \text{ Assoc}(\text{sb}) \Rightarrow (\mu X : \frac{[\text{wrap}, \text{cat}]^\circ}{3} P([\text{tip}, (X \times X) \text{ bin}]) \text{ est}(\text{R})) \sqsubseteq \frac{\text{flatten}^\circ}{3} \text{ est}(\text{R}) \quad (9.3b)$$

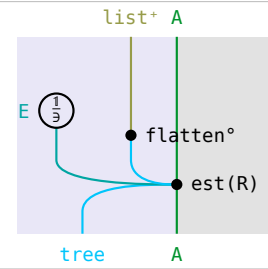
a least-cost bracketing of  $a_1 \oplus \dots \oplus a_n$  is read off an array holding one best tree per non-empty segment, each entry built from the column to its left and the row below it

## circuit

$$\text{list}^+(A) \xrightarrow{\frac{1}{3}} P(\text{list}^+(A)) \xrightarrow{E(\text{flatten}^\circ)} P(\text{tree}(A)) \xrightarrow{\text{est}(\text{R})} \text{tree}(A)$$

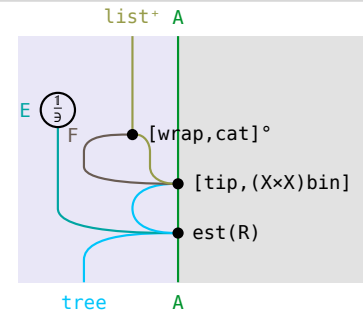
the specification  $\triangleq \text{flatten}$

## Hinze–Marsden



$$\text{list}^+(A) \xrightarrow{\frac{1}{3}} P(\text{list}^+(A)) \xrightarrow{E([\text{wrap}, \text{cat}]^\circ)} PF(\text{list}^+(A)) \xrightarrow{P([\text{tip}, (X \times X) \text{ bin}])} P(\text{tree}(A)) \xrightarrow{\text{est}(\text{R})} \text{tree}(A)$$

split the list in every way, bracket both halves, join. The condition is monotonicity **in context**,  
 $\sqsubseteq F(\text{Rn}(\text{flatten} \text{ flatten}^\circ)) \text{ h} \sqsubseteq \text{hR}$  — only trees with the same flattening are compared — which is Proposition 9.3 at  $\text{H}^\circ = \text{flatten}$ , a map, with (9.5)  $[\text{tip}, \text{bin}] \text{ cost} = (1 + (\text{cost}, \text{flatten})^2) \text{g}$  (the cost of a node reads only the cost and the flattening of its two subtrees) and (9.6)  $(1 + (\leq \times 1)^2) \text{g} \sqsubseteq \text{g} \leq (\text{g} \text{ monotonic on } \leq \text{ in its two cost arguments})$

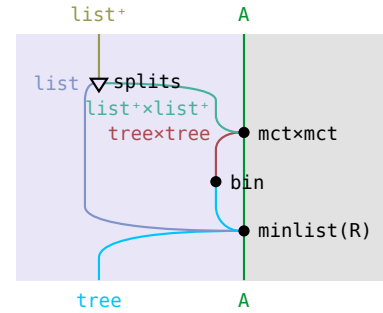


$$\text{list}^+(A) \xrightarrow{\frac{1}{3}} P(\text{list}^+(A)) \xrightarrow{E(\text{cat}^\circ)} P(\text{list}^+(A)^2) \xrightarrow{P((X \times X) \text{ bin})} P(\text{tree}(A)) \xrightarrow{\text{est}(\text{R})} \text{tree}(A)$$

$\equiv$  Proposition 9.1: **wrap** and **cat** have disjoint ranges, and **single** is the coreflexive on singleton lists, where **wrap** returns

$$\text{list}^+(A) \xrightarrow{\text{splits}} [\text{list}^+(A)^2] \xrightarrow{\text{list}((\text{mct} \times \text{mct}) \text{ bin})} [\text{tree}(A)] \xrightarrow{\text{minlist}(\text{R})} \text{tree}(A)$$

$\sqsubseteq \text{splits} \triangleq \langle \text{inits}^+, \text{tails}^+ \rangle$  zip implements  $\frac{\text{cat}^\circ}{3}$  and  $\text{minlist}(\text{R})$  implements  $\text{est}(\text{R})$ . Exponential, since the segments of one list overlap



$$\text{mct} = (\text{single} \rightarrow \text{head tip}, (\text{init col}, \text{tail row}) \text{ mix}) \quad (9.7)$$

$\equiv$  the tabulation: **mct**  $\times$  **s** is needed for every non-empty segment **xs**, so the values are held as an array of rows,  $\text{array} \triangleq \text{inits list}(\text{row}), \text{row} \triangleq \text{tails list}(\text{mct}), \text{col} \triangleq \text{inits list}(\text{mct}), \text{mix} \triangleq \text{zip list}(\text{bin}) \text{ minlist}(\text{R})$

$$\text{col} = (\text{single} \rightarrow \text{head tip wrap}, (\text{init col}, \text{tail row}) \text{ next}) \quad (9.8)$$

$$\text{cons col} = (1 \times \text{array}) \text{ process} \quad (9.9)$$

$$\equiv \text{row} = (\text{single} \rightarrow \text{head tip wrap}, (\text{mct}, \text{tail row}) \text{ cons}) \quad (9.10)$$

a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop,  $\text{next} \triangleq \langle \pi_1, \text{mix} \rangle \text{ snoc}, \text{process} \triangleq ((\text{tip wrap}) \times 1) \text{ loop}(\text{next})$

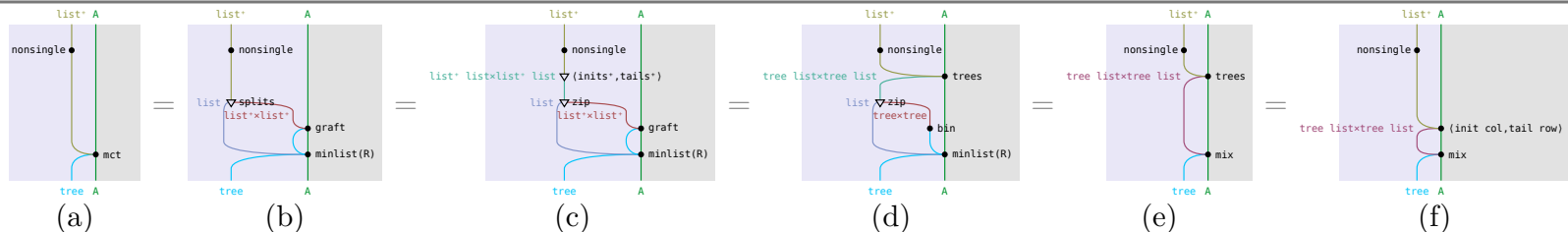
$$\text{array} = ([\text{fstcol}, \text{addcol}]), \text{fstcol} \triangleq \text{tip wrap wrap}$$

$$\equiv \text{addcol} \triangleq \langle \pi_1 \text{ tip wrap}, \text{step} \rangle \text{ cons}, \text{step} \triangleq \langle \text{process tail}, \pi_2 \rangle \text{ zip list}(\text{cons})$$

the program: one fold building the array column by column, cubic in the length of the input

$$\text{nonsingle mct} = \text{nonsingle } (\text{init col}, \text{tail row}) \text{ mix} \quad (9.7)$$

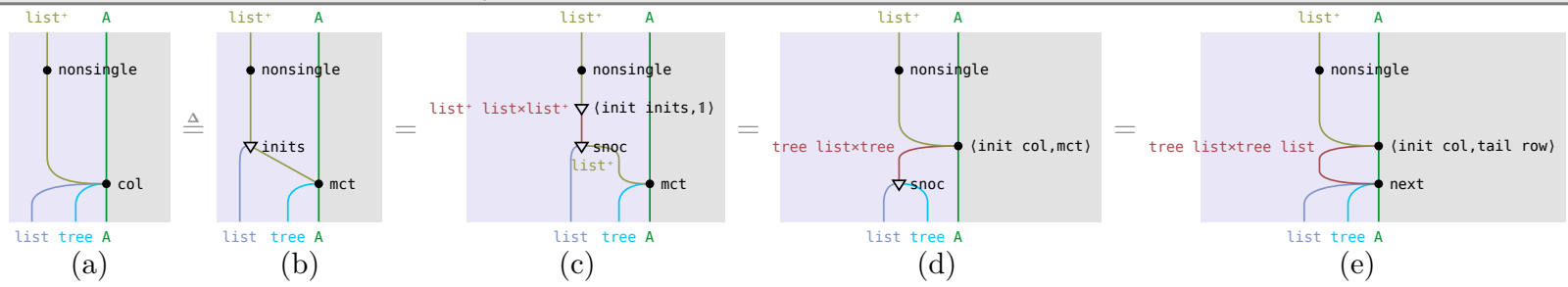
on a list of two or more elements, **mct** gives what **mix** gives on the column of its **init** and the row of its **tail**



|     |                                                                                                                                    |                                                                                                                            |
|-----|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| (a) | nonsingle mct                                                                                                                      | $\text{mct} \triangleq \text{flatten}$                                                                                     |
| (b) | $\equiv \text{nonsingle splits list}(\text{graft}) \text{ minlist}(\text{R})$                                                      | $\text{splits} \triangleq \text{flatten}$                                                                                  |
| (c) | $\equiv \text{nonsingle } \langle \text{inits}^+, \text{tails}^+ \rangle \text{ zip list}(\text{graft}) \text{ minlist}(\text{R})$ | $\text{graft}, \text{trees}; \text{zip list}(f \times g) = (\text{list}(f) \times \text{list}(g)) \text{ zip}$             |
| (d) | $\equiv \text{nonsingle trees zip list}(\text{bin}) \text{ minlist}(\text{R})$                                                     | $\text{mix} \triangleq \text{zip list}(\text{bin}) \text{ minlist}(\text{R}) \triangleq \text{flatten}$                    |
| (e) | $\equiv \text{nonsingle trees mix}$                                                                                                | $\text{inits}^+ = \text{init inits}, \text{tails}^+ = \text{tail tails}; \text{row}, \text{col} \triangleq \text{flatten}$ |
| (f) | $\equiv \text{nonsingle } (\text{init col}, \text{tail row}) \text{ mix}$                                                          |                                                                                                                            |

**nonsingle col = nonsingle (init col,tail row) next**

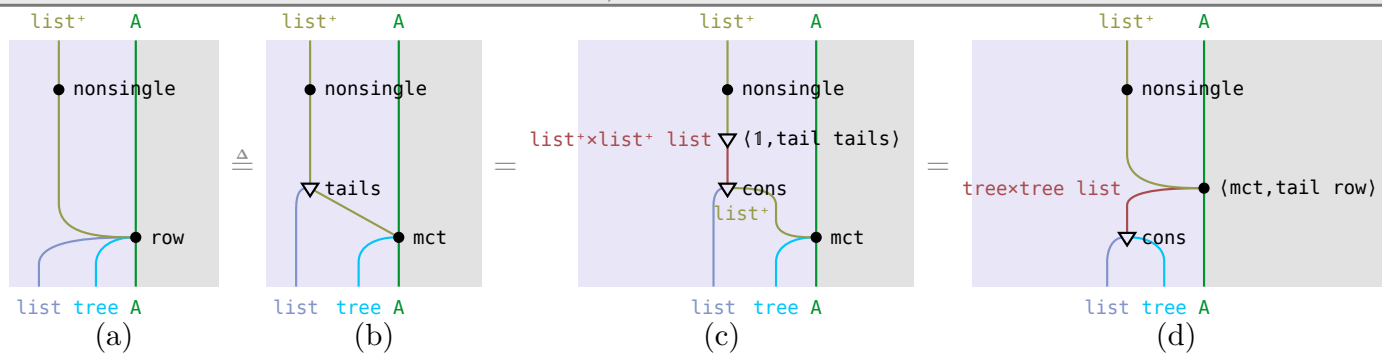
(9.8)

on a list of two or more elements, the column is what **next** makes of the column of its **init** and the row of its **tail**

|     |                                             |                                                                            |
|-----|---------------------------------------------|----------------------------------------------------------------------------|
| (a) | nonsingle col                               | col $\triangleq$ flatten                                                   |
| (b) | $\triangleq$ nonsingle inits list(mct)      | inits=(init inits,1) snoc on non-singletons                                |
| (c) | $=$ nonsingle (init inits,1) snoc list(mct) | snoc list(f)=(list(f) $\times$ f) snoc; col                                |
| (d) | $=$ nonsingle (init col,mct) snoc           | (9.7) — (9.7); next $\triangleq$ ( $\pi_1$ ,mix) snoc $\triangleq$ flatten |
| (e) | $=$ nonsingle (init col,tail row) next      |                                                                            |

**nonsingle row = nonsingle (mct,tail row) cons**

(9.10)

on a list of two or more elements, the row is **mct** of the whole list consed onto the row of its **tail**

|     |                                             |                                             |
|-----|---------------------------------------------|---------------------------------------------|
| (a) | nonsingle row                               | row $\triangleq$ flatten                    |
| (b) | $\triangleq$ nonsingle tails list(mct)      | tails=(1,tail tails) cons on non-singletons |
| (c) | $=$ nonsingle (1,tail tails) cons list(mct) | cons list(f)=(f $\times$ list(f)) cons; row |
| (d) | $=$ nonsingle (mct,tail row) cons           |                                             |

| <b>cons col = (1 × array) process</b> <span style="float: right;">(9.9)</span><br>the column of <b>a</b> consed onto <b>x</b> is what <b>process</b> makes of <b>a</b> and the array of <b>x</b> |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                          | Hinze–Marsden |
|                                                                                                                                                                                                  |               |
|                                                                                                                                                                                                  |               |
|                                                                                                                                                                                                  |               |
|                                                                                                                                                                                                  |               |

| <b>tops = (1 × array) process tail</b> <span style="float: right;">(9.3c)</span><br>the trees at the top of the new rows are what <b>process</b> makes of <b>a</b> and the array of <b>x</b> , less its first |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                       | Hinze–Marsden |
|                                                                                                                                                                                                               |               |
|                                                                                                                                                                                                               |               |
|                                                                                                                                                                                                               |               |
|                                                                                                                                                                                                               |               |

| <b>rests = <math>\pi_2</math> array</b> <span style="float: right;">(9.3d)</span><br>the rest of each new row is the array of <b>x</b> |               |
|----------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                | Hinze–Marsden |
|                                                                                                                                        |               |
| <p>definition of <math>\text{rests} \triangleq \text{tic list}(\text{tail row})</math>, our name</p>                                   |               |
| <p><math>\text{tic list}(\text{tail}) = \pi_2 \text{ inits}</math>; definition of <b>array</b></p>                                     |               |

| <b>newrows = (1 × array) step</b> <span style="float: right;">(9.3e)</span><br>the rows of the array of <b>a:x</b> below its first are what <b>step</b> makes of <b>a</b> and the array of <b>x</b> |               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                             | Hinze–Marsden |
|                                                                                                                                                                                                     |               |
| <p>definition of <math>\text{newrows} \triangleq \text{tic list}(\text{row})</math>, our name</p>                                                                                                   |               |
| <p>(9.10) — (9.10) — on the non-singleton <b>tic</b> lists</p>                                                                                                                                      |               |
| <p><math>\text{list}(f,g) = (\text{list}(f), \text{list}(g))</math> zip; products; definitions of <b>tops</b> and <b>rests</b></p>                                                                  |               |
| <p>(9.3c) and (9.3d); products; definition of <b>step</b></p>                                                                                                                                       |               |

| $\text{cons array} = (1 \times \text{array}) \text{ addcol}$ <span style="float: right;">(9.3f)</span><br>the array of <b>a</b> consed onto <b>x</b> is what <b>addcol</b> makes of <b>a</b> and the array of <b>x</b>                                                                           |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                          | Hinze–Marsden |
|                                                                                                                                                                                                                                                                                                  |               |
| $\text{list}^+(A) \text{ cons } \text{list}^+(A) \text{ inits } [\text{list}^+(A)] \text{ list}(\text{row}) [[\text{tree}(A)]]$<br>definition of <b>array</b>                                                                                                                                    |               |
| $\text{list}^+(A) \text{ cons } \text{list}^+(A) \text{ inits } \text{tail} [\text{list}^+(A)] \text{ cons } [\text{list}^+(A)] \text{ list}(\text{row}) [[\text{tree}(A)]]$<br>$\text{cons inits} = (\pi_1 \text{ wrap, tic}) \text{ cons}$ , abbreviating <b>cons inits tail</b> by <b>tic</b> |               |
| $\text{list}^+(A) \text{ cons } \text{list}^+(A) \text{ wrap } \text{list}^+(A) \text{ row } [\text{tree}(A)] \text{ cons } [[\text{tree}(A)]]$<br>$\text{cons list}(f) = (f \times \text{list}(f)) \text{ cons}$ ; definition of <b>newrows</b>                                                 |               |
| $\text{list}^+(A) \text{ cons } \text{list}^+(A) \text{ tip } \text{tree}(A) \text{ wrap } [\text{tree}(A)] \text{ cons } [[\text{tree}(A)]]$<br>$\text{wrap row} = \text{tip wrap}$                                                                                                             |               |
| $\text{list}^+(A) \text{ array } \text{addcol} [[\text{tree}(A)]]$<br>(9.3e); products; definition of <b>addcol</b>                                                                                                                                                                              |               |

## §9.4. Data compression

(9.4a)

$[A] ::= \text{nil} \mid \text{snoc } ([A], A), \text{ list}^+ A ::= \text{wrap } A \mid \text{snoc } (\text{list}^+ A, A), \text{ String} = [\text{Char}], \text{ Code} ::= \text{sym } \text{Char} \mid \text{ptr } (\text{String}, \text{String}^+), F(\text{Code}, \text{String}) = 1 + (\text{String} \times \text{Code}), \alpha \triangleq [\text{nil}, \text{snoc}].$

$\text{decode} \triangleq ([\text{nil}, \text{extend}]) : [\text{Code}] \rightarrow \text{String}, \text{ extend } (xs, \text{sym } a) = xs \# [a], \text{ extend } (xs, \text{ptr } (ys, zs)) = xs \# zs \text{ when } ys \# zs \text{ is a proper prefix of } xs \# zs; H = \text{decode}^\circ.$

$\text{size} \triangleq ([\text{zero}, \text{distr } [1 \times c, 1 \times p] \text{ plus}])$  with  $c, p$  the constant costs of a symbol and a pointer;  $R \triangleq \text{size} \leq \text{size}^\circ, Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau))$ , the two  $\tau$  on symbols and on pointers.

$\text{lrt } ws = \text{est}(\text{prefix}^\circ \times (\tau + \tau)) \{(xs, (ys, zs)) \mid xs \# zs = ws, ys \# zs \text{ a proper prefix of } ws\}$ , the longest repeated tail;  $\text{reduce } (ws \# [a]) = (zs \neq [] \rightarrow [(ws, \text{sym } a), (xs, \text{ptr } (ys, zs))], [(ws, \text{sym } a)])$  with  $(xs, (ys, zs)) = \text{lrt } (ws \# [a]).$

$$0 \leq c \wedge 0 \leq p \Rightarrow (\mu X : \frac{[\text{nil}, \text{extend}]^\circ}{3} \text{thin}(Q) P([\text{nil}, (X \times 1) \text{snoc}]) \text{est}(R)) \sqsubseteq \frac{\text{decode}^\circ}{3} \text{est}(R), \quad (9.4b)$$

$$\text{encode} = (\text{null} \rightarrow \text{nil}, \text{reduce list}((\text{encode} \times 1) \text{snoc}) \text{minlist}(R))$$

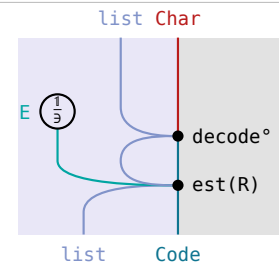
a smallest code sequence decoding to the given string is built from the right, each step emitting the last character as a symbol or ending with a pointer back into what has already been decoded

## circuit

$$[\text{Char}] \xrightarrow{\frac{1}{3}} P[\text{Char}] \xrightarrow{E(\text{decode}^\circ)} P[\text{Code}] \xrightarrow{\text{est}(R)} [\text{Code}]$$

the specification  $\dashv\vdash \text{decode}$

## Hinze–Marsden

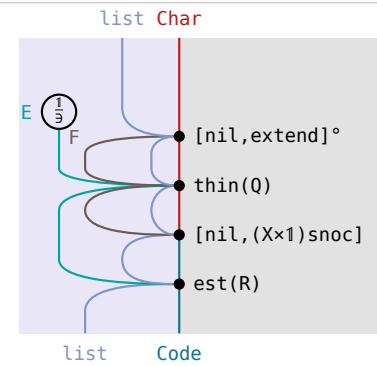


$$[\text{Char}] \xrightarrow{\frac{1}{3}} P[\text{Char}] \xrightarrow{E([\text{nil}, \text{extend}]^\circ)} \text{thin}(Q) \xrightarrow{P[\text{Char}]} P([\text{nil}, (X \times 1) \text{snoc}]) \xrightarrow{P[\text{Code}]} \text{est}(R) \xrightarrow{[\text{Code}]}$$

at  $Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau))$ , the two  $\tau$  on symbols and on pointers.

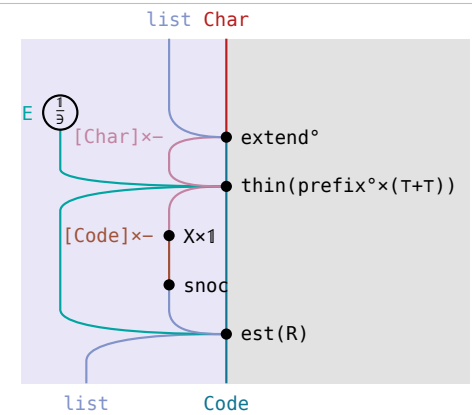
Monotonicity  $F(R) \alpha \sqsubseteq \alpha R$  is routine, the two costs being constants:  $(\tau + \tau)[c, p] = [c, p]$ .

Proposition 9.4 splits the thinning condition in two —  $F(\tau + \tau, R) \alpha \sqsubseteq \alpha R$ , and  $\text{decode prefix} \sqsubseteq R^\circ \text{decode}$ , for which  $\text{decode init} \sqsubseteq R^\circ \text{decode}$  is enough: dropping the last character of the output shortens or removes the last code element and never raises the cost. So between a symbol and a pointer nothing can be decided in advance, and between two pointers the longer match wins



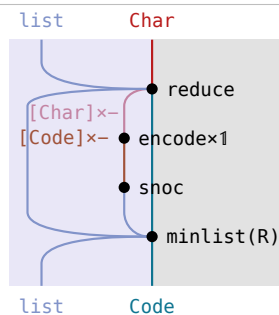
$$[\text{Char}] \xrightarrow{\frac{1}{3}} P[\text{Char}] \xrightarrow{E(\text{extend}^\circ)} \text{thin}(\text{prefix}^\circ \times (\tau + \tau)) \xrightarrow{P([\text{Char}] \times \text{Code})} P((X \times 1) \text{snoc}) \xrightarrow{P[\text{Code}]} \text{est}(R) \xrightarrow{[\text{Code}]}$$

Proposition 9.1:  $\text{nil}$  and  $\text{extend}$  have disjoint ranges. The decompositions of one string are  $\frac{\text{extend}^\circ}{3} (ws \# [a]) = \{(ws, \text{sym } a)\} \cup \{(xs, \text{ptr } (ys, zs)) \mid xs \# zs = ws \# [a], ys \# zs \text{ a proper prefix of } ws\}$  — take the last character as a symbol, or end with a pointer



$$[\text{Char}] \xrightarrow{\text{reduce}} [[\text{Char}] \times \text{Code}] \xrightarrow{\text{list}((\text{encode} \times 1) \text{snoc})} [[\text{Code}]] \xrightarrow{\text{minlist}(R)} [\text{Code}]$$

$\text{reduce}$  implements  $\frac{\text{extend}^\circ}{3} \text{thin}(\text{prefix}^\circ \times (\tau + \tau))$ : thinning leaves at most two, the symbol and the pointer of the longest repeated tail  $\text{lrt}$ . Again exponential; the book gives no tabulation for it

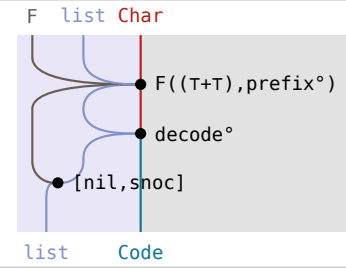
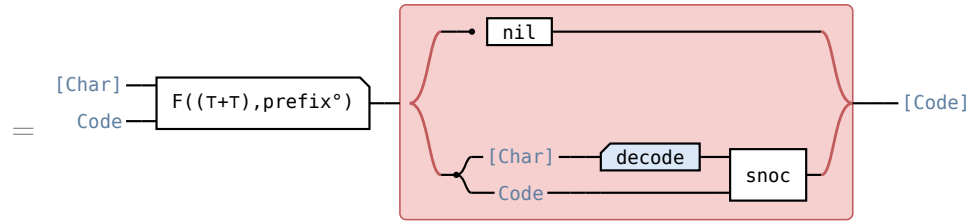
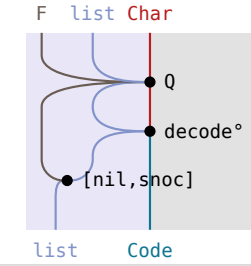
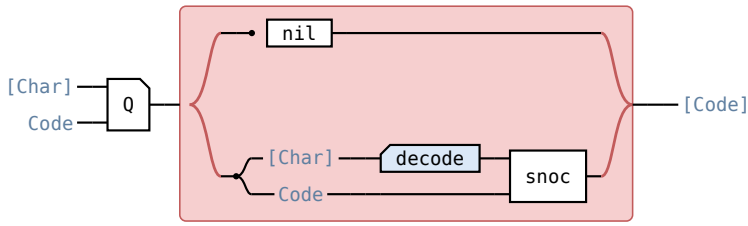


$$\theta \leq c \wedge \theta \leq p \Rightarrow Q \text{ F}(\text{decode}^\circ) [\text{nil}, \text{snoc}] \sqsubseteq \text{F}(\text{decode}^\circ) [\text{nil}, \text{snoc}] R \quad (9.4c)$$

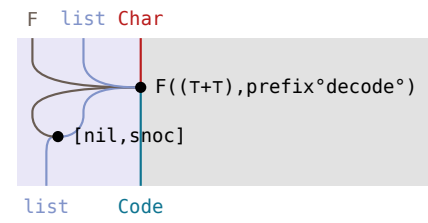
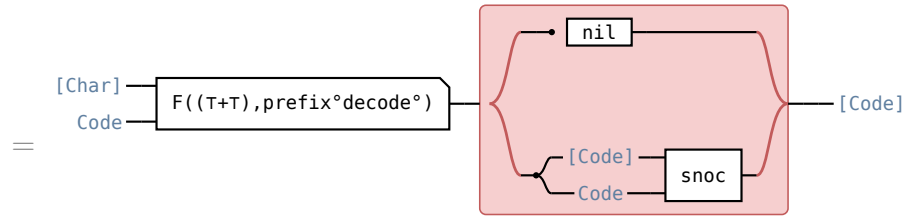
every code sequence built from a decomposition  $Q$  puts above a given one — a code element of the same kind, a longer front string — costs at least as much as one built from the given one

**circuit**

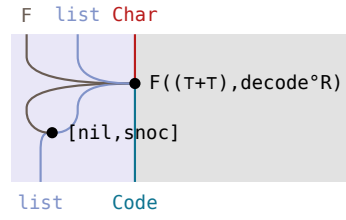
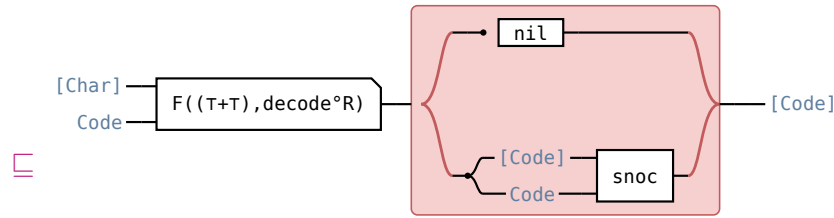
**Hinze–Marsden**



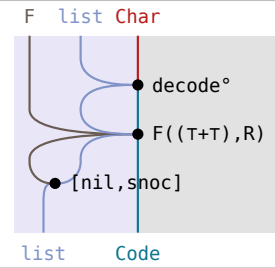
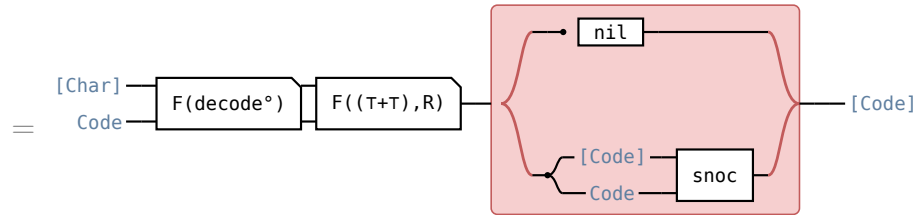
$Q \triangleq 1 + (\text{prefix}^\circ \times (T+T))$  is  $F(T+T, \text{prefix}^\circ) \triangleq \text{decode}$



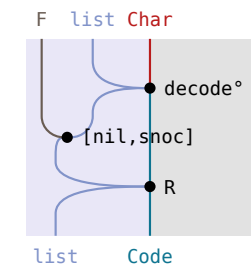
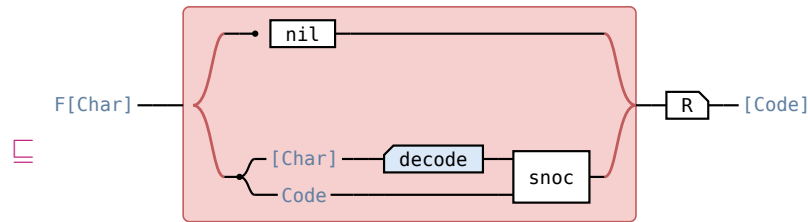
the bifunctor  $F$  preserves composition:  $F(U, \text{prefix}^\circ) F(1, \text{decode}^\circ) = F(U, \text{prefix}^\circ \text{decode}^\circ)$



Proposition 9.4's second condition  $\text{prefix}^\circ \text{decode}^\circ \sqsubseteq \text{decode}^\circ R$ : dropping the last **sym**, or shortens or drops the last pointer. Proved at **prefix** directly by induction, not through **init**



$F$  preserves composition:  $F(U, \text{decode}^\circ R) = F(1, \text{decode}^\circ) F(U, R)$



Proposition 9.4's first condition  $F(T+T, R) \alpha \sqsubseteq \alpha R$ , left as an exercise in the book: **snoc** adds the cost of the last element to both sides, and  $[c, p](T+T) = [c, p]$

| <div> <div>extend nil<sup>°</sup> = 0</div> <div>(9.4d)</div> </div> <div>no string is both a result of <b>extend</b> and the result [] of <b>nil</b></div> |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                     | Hinze–Marsden |
|                                                                                                                                                             |               |
| <p>nil returns only [], so nil<sup>°</sup>=null nil<sup>°</sup></p>                                                                                         |               |
| <p>extend null=0: a symbol ends the string with a character, a pointer with its non-empty ZS</p>                                                            |               |
| <p>0 composed with anything is 0</p>                                                                                                                        |               |